Distance of RAA Codes over Large Finite Fields (with Applications in zkSNARKs and PCGs)

Pariya Akhiani and Yupeng Zhang

University of Illinois Urbana Champaign
{akhiani2,zhangyp}@illinois.edu

Abstract. Error-correcting codes play a central role in modern cryptography, enabling efficient constructions of primitives such as zero knowledge proofs and secure multiparty computations. Among them, repeat-accumulate-accumulate (RAA) codes have recently attracted significant attention due to their linear-time encoding and good distance properties. However, prior work only established provable distance guarantees over the binary field or over finite fields whose size is smaller than the message length. The underlying techniques do not extend to larger fields. This restriction is significant, as many cryptographic constructions based on error-correcting codes operate over large finite fields.

In this paper, we prove that RAA codes achieve constant relative distance $0 < \delta \leq \frac{1}{2}$ with high probability over large finite fields. Moreover, we resolve an open conjecture by showing for the first time that the distance of RAA codes improves over large fields. We provide both theoretical analysis and empirical evidence demonstrating that, compared to the binary case, large fields yield strictly better distance guarantees with much smaller failure probabilities.

Our empirical evaluation shows that for message length $n = 2^{15}$ and repetition factor $r = 4$, the RAA code over a 31-bit prime field achieves relative distance $1/2$, except with failure probability 2^{-16}. This is better than the best provable distance of $\delta = 0.2$ with failure probability 2^{-7} over the binary field. Leveraging our improved distance bounds, we obtain a $2.6\times$ reduction in the proof size of a polynomial commitment scheme based on RAA codes compared to using the binary-field bound, and a $5.4\times$ reduction compared to a prior construction based on expand-accumulate codes.

1 Introduction

Error-correcting codes play an important role in many constructions of cryptographic primitives. For example, they are used to build pseudorandom correlation generators (PCGs), which are powerful tools to generating correlated randomness in oblivious transfers (OT) and vector oblivious linear evaluations (VOLE) with sublinear communication cost. There is also an important category of zero-knowledge succinct non-interactive arguments of knowledge (zkSNARKs) based on error-correcting codes, with advantages of efficient prover time, transparent setup and plausibly post-quantum security. In these schemes, efficiency is largely determined by the encoding time of the underlying error-correcting code, whereas security relies on the distance of the code. Consequently, developing error-correcting codes with efficient encoding and good distance has become a central research focus in these areas.

The *Repeat–Accumulate–Accumulate* (RAA) code has recently regained attention due to its linear-time encoding. In [BCF+25], RAA code is proven to achieve good distance both asymptotically and concretely, and is used to construct an efficient zkSNARK scheme called Blaze. However, the proof is only for RAA codes over the binary field, and does not generalize to larger finite fields, which are of great interest in cryptographic applications. As a consequence, many constructions utilizing RAA codes over finite fields such as [BFRW25, BMMS25] only have heuristic security[1]. In fact, Blaze [BCF+25] has to be constructed over a binary extension field due to this limitation, unlike most zkSNARKs based on error-correcting codes that are over large finite fields.

It is widely conjectured that the distance of this family of codes improves over larger fields [BCG+22]. However, no work in the literature has formally proven this conjecture. This is because analyzing the distance over a large finite field is technically challenging, as techniques such as Markov's inequality and the naive union bound, which are effective in the binary setting, no longer work. Almost all existing papers in the literature on this family of error-correcting codes based on the accumulator matrix focus on the binary field. The only exception is [BFK+24], where the authors provided a formal proof that the expand-accumulate (EA) code has good distance over any field with high probability. Unfortunately, the provable distance bound is even weaker than the binary case due to limitations of the proof techniques, contradicting the conjecture that distance should improve as the field size increases.

Our Contributions. In this paper, we solve this open problem by formally proving that the RAA code over a large finite field achieves good distance with high probability. More importantly, we show – both theoretically and through concrete parameter instantiations – that our provable distance is strictly better than that of the binary-field counterpart, a result that has not been achieved in any prior work. Specifically, our contributions are summarized as follows:

– We show that an RAA code with message length n, repetition factor r and codeword length $N = rn$ over a finite field of size q has constant relative distance $\delta \leq \frac{1}{2}$, except with a small failure probability that is inverse polynomial in N. The dominating terms of the failure probability are $\tilde{O}(N^{2-r}) + O(\frac{1}{q-1})$, where $\tilde{O}(\cdot)$ hides polylogarithmic factors. This failure probability is strictly smaller than that in the binary case, which is $O(N^{1+\gamma-\frac{r}{2}})$ for a constant $0 < \gamma < 1$, as shown in [BCF+25]. Moreover, our provable distance bound is also strictly better than the binary-field bound, which only guarantees $\delta < \frac{1}{3}$. Our key technical contribution is a fine-grained analysis of the accumulator

[1] The focus of these papers is on the Interactive Oracle Proof of Proximity (IOPP), which works with any error-correcting code, but RAA code is used to achieve linear prover time.

matrix over large fields. This technique may be of independent interest and can be applied to strengthen the distance analysis of other codes in this family over finite fields, such as the EA code.

- As a by-product, our proof strategy can also be used to improve the analysis of the RAA code over the binary field. In particular, we reduce the dominating term of the failure probability to $\tilde{O}(N^{1-\frac{r}{2}})$, which may be of independent interest. We note that this improved failure probability remains worse than the failure probability achieved over large finite fields.
- Finally, we implement Sage scripts to compute the provable distance and corresponding failure probability under various parameter settings. Our evaluation shows that for $n = 2^{15}$, $r = 4$, the RAA code over a 31-bit prime field achieves a relative distance of $\delta = \frac{1}{2}$, except with failure probability 2^{-16}. This significantly improves upon the binary setting, which guarantees only $\delta = 0.2$ with failure probability 2^{-7}.
  To demonstrate the practical implications of our improved bounds, we evaluate the proof size of a polynomial commitment scheme (PCS) constructed from error-correcting codes, a fundamental building block of many zkSNARK systems. Leveraging our tighter distance bound, the resulting proof size for a polynomial of size $m = 2^{20}$ is 0.48 MB. This is 2.6× smaller compared to using the provable distance of the RAA code in [BCF+25] over a binary extension field, and is 5.4× smaller compared to the PCS in [BFK+24], which is based on the EA code over a finite field.

Applications in Cryptography. Our results have important applications in cryptography. As mentioned above, zkSNARKs based on error-correcting codes have drawn great attention recently. The constant relative distance is critical to prove their security through the proximity gap, while the encoding time dominates the prover efficiency. Most schemes are over large finite fields, due to the use of randomness and the Schwartz-Zippel Lemma [Sch80, Zip79] in the constructions. The only other candidate error-correcting code with linear-time encoding and constant relative distance is the (generalized) Spielman code [Spi95], but its concrete distance is very poor in practice ($\delta = 0.07$ for $n = 2^{30}$ [GLS+23]), leading to large proof size of tens of megabytes. Our result provides another option to build zkSNARKs with linear prover time over any finite field using the RAA code, while both the prover time and the proof size are better than those using the Spielman code.

Another important application is constructing PCGs based on error-correcting codes with sublinear communication. The large distance of the code is crucial to establish their security based on variants of Learning Parity with Noise (LPN) assumptions, where almost all known attacks fit in the framework of finding low-weight codewords [BCG+22]. Though RAA code has not been directly used in PCG constructions, other codes in the same family such as the EA code [BCG+22] and the Expand-Convolute code [RRT23] have been used, and the RAA code can serve as a drop-in replacement to improve the computation time to linear over large finite fields. Overall, our results will enable new constructions with better performance in both areas.

	Field	Asymptotic Failure Probability	δ	Concrete Failure Probability
[BCF+25]	$\mathbb{F}_2$	$O(N^{1+\gamma-\frac{r}{2}})$	0.2	2^{-7}
[AHKM25]	$\mathbb{F}_q$	-	0.05	2^{-21}
Our result	$\mathbb{F}_q$	$\tilde{O}(N^{2-r})$	0.5	2^{-16}

Table 1: Comparison with existing works on the distance of RAA code. "-" means not explicitly derived in the paper. The concrete δ and failure probability are for $n = 2^{15}, r = 4, q > 2^{31}$.

Moreover, cryptographic applications operate in parameter regimes that differ substantially from those traditionally studied in coding theory. In particular, for security reasons, the finite field size q is typically asymptotically larger than any polynomial in n, a regime that has not been carefully analyzed in the coding theory literature. For example, in [RA11], the authors claim that the RAA code is asymptotically good over finite fields. However, their proof assumes that $n >> q$ as $n \to \infty$, and thus the result follows in a straightforward manner from the analysis without the size of the field. Our work is the first to show that the RAA code achieves good distance over the large finite fields required by cryptographic applications.

1.1 Related Works

Binary field. The family of repeat–multiple–accumulate (RMA) codes was proposed and analyzed in several early works [DP97, DJM98, PS03, BMS08, KZC07, KZKC08, FR08]. It was shown that RA codes with a single accumulator matrix are not asymptotically good, whereas codes with more than one accumulator matrices, such as RAA codes, achieve constant relative distance with high probability as $n \to \infty$. These works primarily focus on asymptotic behavior, and the concrete parameters obtained in the proofs are not practical. More recently, RAA codes were carefully analyzed in [BCF+25] to obtain improved concrete parameters and were used to construct a SNARK called Blaze. All of these results are over the binary field, and the techniques do not generalize easily to larger finite fields. Moreover, even over the binary field, the parameters remain suboptimal. In particular, the failure probability is $O(N^{1+\gamma-\frac{r}{2}})$, which exceeds $O(N^{-1})$ when $r = 4$. As a result, the code is only effective for large n, and additional techniques – such as puncturing and explicitly checking all messages of weight 1 and 2 – must be introduced to boost the success probability.

A variant called the EA code was introduced in [BCG+22], which replaces the first repeat-accumulate stage of the RAA code with an expander matrix. The work proves that EA codes achieve good distance over the binary field and uses them to construct a PCG. However, the encoding time of the EA code increases to $O(n \log n)$.

Finite field. The good-distance property of RMA codes was extended to finite fields in [Yan04, KCL11, RA11, AR11] under the name weighted non-binary RMA codes. However, some parts of their proofs assume that q is fixed and that n can grow much larger than q as $n \to \infty$, which does not match the parameter regime relevant to cryptographic applications. In [BFK+24], the authors provided the first formal analysis of the EA code over large finite fields with $q \gg n$ and proved that the EA code achieves good distance. This technique was later generalized to RAA codes over large finite fields in [AHKM25], where the authors presented numerical estimates of the failure probability but did not derive asymptotic bounds. Across all of these works, the provable failure probabilities are strictly worse than those obtained over the binary field, which contradicts the natural conjecture that the distance should improve as the field size increases.

A comparison with related works with concrete numbers is in Table 1. For [BCF+25], we wrote a script to compute the exact failure probability using the equation in the paper, without additonal testing and puncturing. For [AHKM25], we use the numbers reported in the paper. As shown in the table, our analysis results in much better distance with smaller failure probability. See Section 5 for more experimental results.

1.2 Technical Overview

The encoding of an RAA code is defined by a repetition code replicating each element of the message r times, followed by two accumulators computing the prefix-sum of their input vectors. The input vector is permuted before each accumulator, and the sampling of the permutation matrices defines a specific RAA code and is the random space of the probability whether the distance of the code is good or not. When defined over a large finite field, each permutation matrix $\mathbf{P}$ can be viewed as the product of a binary matrix $\mathbf{\Pi}$ permuting the positions of elements and a diagonal matrix $\mathbf{V}$ randomizing the values of elements. See Section 2.2 for the formal definition of the RAA code.

Fine-grained analysis of IOWE. Our starting point is the input-output-weight-enumerator (IOWE) of the accumulator matrix, which counts the number of distinct input vectors of weight w that lead to output vectors of weight h. As shown in [AR11, BFK+24], the formula over a finite field of size q is

$$A_{w,h}^{N,q} = \sum_{i=0}^{w-1} \binom{h-1}{\lceil \frac{w-i}{2} \rceil - 1} \binom{N-h}{\lfloor \frac{w-i}{2} \rfloor} \binom{h - \lceil \frac{w-i}{2} \rceil}{i} (q-1)^{\lceil \frac{w-i}{2} \rceil} (q-2)^i,$$

where N is the size of the vector as well as the length of the final codeword. The IOWE has been used to analyze the distance of the RAA code in many prior works, as the probability of an input vector of weight w outputting a vector of h after accumulation can be computed as $\frac{A_{w,h}^{N,q}}{\binom{N}{w}(q-1)^w}$, over the random choice of the permutation matrix before accumulation. With this probability, one could upper-bound the probability of the existence of a "bad" codeword of weight $w_3 < \delta N$ by the union bound summing over all weights of nonzero messages

$w_0 \in [1, n]$, intermediate weights $w_2 \in [1, N]$ and output weights $w_3 \in [0, \delta N)$. Unfortunately, applying the union bound naively does not work in the finite field case. This is because the total number of possible messages of weight w_0 is $\binom{n}{w_0}(q-1)^{w_0}$. When multiplied by the probability above for $w = w_2$ and $h = w_3$, the upper bound becomes exponential in $(q-1)$ for some w_0, w_2, w_3 in their ranges, and is thus meaningless.

In this paper, we propose a fine-grained analysis showing that for all messages of weight w_0, conditioning on the weight of the intermediate vector before the accumulator matrix being w_2, the probability that there exists at least one output vector of weight w_3 after the accumulator matrix is upper-bounded by

$$\binom{n}{w_0} \sum_{i=w_2-2w_0-1}^{w_2-1} \frac{\binom{w_3-1}{\lceil \frac{w_2-i}{2} \rceil - 1}\binom{N-w_3}{\lfloor \frac{w_2-i}{2} \rfloor}\binom{w_3-\lceil \frac{w_2-i}{2} \rceil}{i}}{\binom{N}{w_2}} + O(\frac{1}{q-1}).$$

The analysis consists of union bounds in two steps, one on the random permutation of positions defined by $\mathbf{\Pi}$ and the other on the random values defined by $\mathbf{V}$. See Section 3.1 for the formal statement and the rigorous proof.

Comparing to the approaches in prior works, the upper-bounds in [AR11, BFK+24, AHKM25] essentially drop the terms related to $q-1$, resulting in the same summand with the combinatorial terms above but with i from 0 to $w_2 - 1$. Because of this, the bound is strictly larger than that of the binary case, as the latter consists of a single term of the summand with $i = 0$. It makes the provable bound worse than that of the binary case, as we are bounding the failure probability that there exists a "bad" codeword. We show that our fine-grained analysis is critical to address this issue, making the bound better than that of the binary case.

Good distance with amplification in stage 1. We then divide the analysis into two stages, one for the repetition and the first accumulator (i.e., the RA code), and the other for the second accumulator. We show that if the first stage can amplify the weight of the message from w_0 to $w_2 = O(w_0 \log N)$, then the weight of the codeword w_3 after the second accumulator must be $\geq \delta N$, except for a small probability that is $O(N^{-\eta_1}) + O(\frac{1}{q-1})$. This is inverse polynomial in N and can be made arbitrarily small depending on the constant of the amplification of w_2. See Section 3.2 for the formal statement and the proof. Moreover, we show in Appendix A.1 that our δ can be much larger than the provable distance of the binary case using this approach, while the failure probability is significantly smaller than that of the binary case.

Amplification of stage 1. After that, we show that the first stage of the RA code indeed gives the desired amplification with high probability. That is, for all w_0 up to $O(\frac{n}{\log^2 n})$, $w_2 = O(w_0 \log N)$ after the first accumulator, except for a small probability that is $\tilde{O}(N^{2-r}) + O(\frac{1}{q-1})$, where r is the repetition factor and is thus the inverse of the rate of the code. The failure probability is again significantly smaller than that of the binary case, which is $O(N^{1+\gamma-\frac{r}{2}})$ for a constant $0 < \gamma < 1$ as shown in [BCF+25]. Note that although the analysis in [BCF+25] also consists of two stages, our technique is completely different as we deal with the case of large finite fields as explained above. See Section 3.3 for the details.

High-weight messages. Combining the two stages, so far we can show that the codewords of all messages of weight up to $O(\frac{n}{\log^2 n})$ are "good" with high probability, which we call low-weight messages. We note that the low-weight case must exist in our proof strategy, as r is a constant and $N = rn$, thus it is impossible to amplify the weight for messages with $w_0 = \omega(\frac{n}{\log n})$ in stage 1. We are only able to prove the amplification for $w_0 = O(\frac{n}{\log^2 n})$. Nevertheless, we complete the proof of high-weight messages using the existing approach in [RA11], which builds on top of the proof in [AR09] and [KZKC08]. While no explicit parameters are given in these prior works, for the case of w_0 larger than $O(\frac{n}{\log^2 n})$, the failure probability that there exists a low-weight codeword for any high-weight message is negligible in N. See Section 4 for the details.

2 Preliminaries

We use $\mathbb{F}_q$ to denote a finite field of size q. λ denotes the security parameter, and $\mathsf{negl}(\lambda)$ denotes a negligible function in λ. For positive integer n, we define $[n] := \{0, \ldots, n-1\}$. We use boldface lowercase letters for vectors and boldface uppercase letters for matrices. Vectors are row vectors by default. For a vector $\mathbf{v}$, $\mathsf{wt}(\mathbf{v})$ denotes the Hamming weight (i.e., number of nonzero entries) of $\mathbf{v}$. For two vectors $\mathbf{u}, \mathbf{v}$ of the same dimension, $\Delta(\mathbf{u}, \mathbf{v}) = \mathsf{wt}(\mathbf{u} - \mathbf{v})$ denotes their Hamming distance.

2.1 Error-Correcting Codes

A linear error-correcting code of codeword length N and message length $n \leq N$ over a field $\mathbb{F}_q$ is an n-dimensional linear subspace $C \subseteq \mathbb{F}_q^N$. Such a code is defined by a rank-n generator matrix $\mathbf{G} \in \mathbb{F}_q^{n \times N}$; the encoding of a message $\mathbf{x} \in \mathbb{F}_q^n$ is given by $\mathbf{x} \cdot \mathbf{G} \in \mathbb{F}_q^N$. The *rate* of the code is $R = n/N$. The *distance* of the code C is the minimum Hamming distance between any two codewords in C. When C is a linear code, the minimum distance is equal to the minimum weight of any nonzero codeword $\min_{\mathbf{u} \in C, \mathbf{u} \neq \mathbf{0}} \mathsf{wt}(\mathbf{u})$. We let $\delta := \min_{\mathbf{u} \in C, \mathbf{u} \neq \mathbf{0}} \mathsf{wt}(\mathbf{u})/N$ denote the relative distance of C. We say that a code (or formally a family of code $\{C_n\}$ for different n) is asymptotically good if R and δ remain constants as $n \to \infty$.

2.2 Repeat-Accumulate-Accumulate Code

The encoding matrix of the RAA code of message length n and codeword length N consists of three types of matrices defined over $\mathbb{F}_q$:

- Repetition matrix $\mathbf{R} \in \mathbb{F}_q^{n \times N}$: the matrix repeats each entry of the message vector $r = \frac{N}{n}$ times. The rate of the RAA code is thus $R = \frac{1}{r}$.

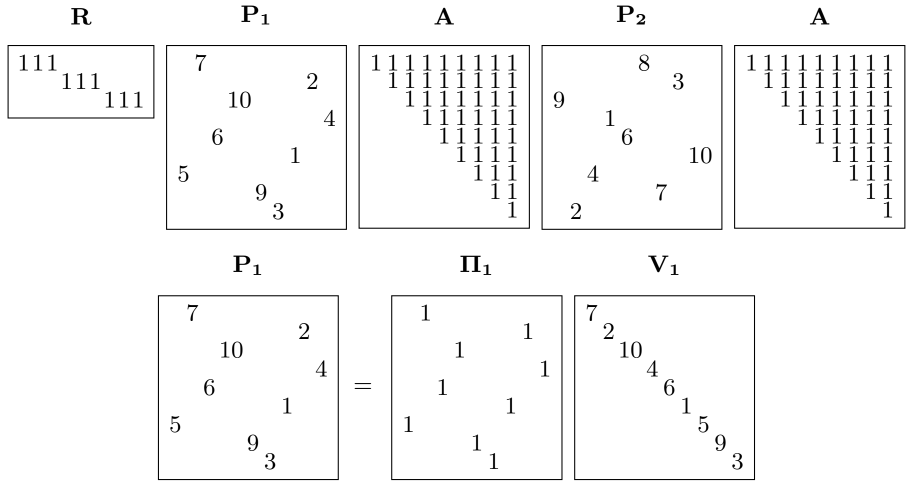

Fig. 1: A pictorial representation of the generator matrix for an RAA code over $\mathbb{F}_{11}$ with rate $r = 1/3$ and codeword length $N = 9$.

– Permutation matrix $\mathbf{P} \in \mathbb{F}_q^{N\times N}$: the matrix randomly permutes the entries of the input vector. When defined over a large finite field, the matrix also multiplies each entry with a random nonzero value over the finite field. In our analysis, it is easier to view this permutation as the product of two $N \times N$ matrices $\mathbf{P} = \mathbf{\Pi} \cdot \mathbf{V}$, where $\mathbf{\Pi}$ is a permutation matrix with binary values for permutation $\pi : [N] \to [N]$ such that $\mathbf{\Pi}[i, j] = 1$ if and only if $\pi(i) = j$; $\mathbf{V}$ is a randomization matrix with nonzero values over the diagonal sampled uniformly from $\mathbb{F}_q$. We sometimes use $\mathbf{P}$ for short in our analysis.
– Accumulator matrix $\mathbf{A} \in \mathbb{F}_q^{N\times N}$: the matrix computes the prefix-sum of the input vector, hence the name “accumulator”. $\mathbf{A}[i, j] = 1$ if and only if $i \le j$.

With these matrices, the generator matrix of the RAA code is defined as

$$\mathbf{G} = \mathbf{R} \cdot \mathbf{\Pi}_1 \cdot \mathbf{V}_1 \cdot \mathbf{A} \cdot \mathbf{\Pi}_2 \cdot \mathbf{V}_2 \cdot \mathbf{A},$$

where $\mathbf{\Pi}_1$ and $\mathbf{\Pi}_2$ are two permutations of length N sampled uniformly at random, and $\mathbf{V}_1$ and $\mathbf{V}_2$ are $N \times N$ diagonal matrices with diagonal entries sampled uniformly from $\{1, 2, \ldots, q-1\}$. An example of the generator matrix is shown in Figure 1.

Though RAA codes with $r = 2$ can be asymptotically good [BMS08], $r \ge 3$ is required for good concrete parameters [KZC07, KZKC08]. In this paper, we consider the case of $r \ge 4$ for the convenience of our proof, and we believe our results can be generalized to the case of $r = 3$ with careful analysis.

In our analysis in the paper, we use w_0 to denote the weight of a specific message $\mathbf{x}$, $w_1 = rw_0$ to denote the weight of the vector after the repetition matrix $\mathbf{R}$, w_2 as the weight after the first accumulator matrix $\mathbf{A}$, and w_3 as the weight after the second accumulator matrix, which is the weight of the output codeword. The notations are summarized in Table 2.

2.3 IOWE of Accumulator Matrix A

The *input-output-weight-enumerator (IOWE)* is an important technique used to analyze the distance of the family of RAA codes in the literature. It counts the exact number of input vectors of weight w that lead to output vectors of weight h for a fixed matrix. The IOWE of the accumulator matrix $\mathbf{A}$ over a finite field $\mathbb{F}_q$ is given by the following theorem:

Theorem 1 ([AR11, BFK⁺24]). *The IOWE of the accumulator matrix* $\mathbf{A} \in \mathbb{F}_q^{N\times N}$ *with input weight* w *and output* h *is:*

$$A_{w,h}^{N,q} = \sum_{i=0}^{w-1} \binom{h-1}{\lceil\frac{w-i}{2}\rceil - 1}\binom{N-h}{\lfloor\frac{w-i}{2}\rfloor}\binom{h-\lceil\frac{w-i}{2}\rceil}{i}(q-1)^{\lceil\frac{w-i}{2}\rceil}(q-2)^i. \tag{1}$$

We refer the readers to [AR11, BFK⁺24] for the full proof of the theorem. A simple fact about the output weight is that $h \geq \lceil\frac{w}{2}\rceil$, i.e., the output weight cannot be less than half of the input weight after the accumulator matrix (see [BFK⁺24, Lemma 2.1] for a proof). Moreover, the third combinatorial term implies that $i \leq h - \lceil\frac{w-i}{2}\rceil$, or equivalently $i \leq 2h - w + 1$. That is, h can be as small as $\lceil\frac{w}{2}\rceil$, but in the range of the summation $0 \leq i \leq w-1$, when $2h - w + 1 < w - 1$, the terms in the summand of Equation (1) for $2h - w + 1 < i \leq w - 1$ are all 0, following the convention that $\binom{a}{b} = 0$ if $b > a$.

Here we define a property of the IOWE of $\mathbf{A}$. In Equation (1), for each i, the three combinatorial terms in the summand and the two terms related to q are separable. That is, for a specific $i \in [w]$, there are exactly $\binom{h-1}{\lceil\frac{w-i}{2}\rceil-1}\binom{N-h}{\lfloor\frac{w-i}{2}\rfloor}\binom{h-\lceil\frac{w-i}{2}\rceil}{i}$ possible positions to place the nonzero values in the input vector that could lead to output of weight h, out of $\binom{N}{w}$ possible positions; in each of these positions,

Symbol	Description
r	repetition factor
n	message length
N	codeword length, $N = rn$
$\mathbf{x}$	input vector $\in \mathbb{F}_q^n$
$\mathbf{R}$	repetition matrix $\in \mathbb{F}_q^{n\times N}$
$\mathbf{A}$	accumulator matrix, $\in \mathbb{F}_q^{N\times N}$
w_0	input weight, $\mathsf{wt}(\mathbf{x})$
w_1	weight after repetition, $\mathsf{wt}(\mathbf{xR}) = rw_0$
w_2	weight after the first accumulation, $\mathsf{wt}(\mathbf{xRP}_1\mathbf{A})$
w_3	output weight, $\mathsf{wt}(\mathbf{xRP}_1\mathbf{AP}_2\mathbf{A})$

Table 2: Notation.

there are $(q-1)^{\lceil \frac{w-i}{2} \rceil}(q-2)^i$ values that indeed have output weight h, out of $(q-1)^w$ possible values in total.

Formally speaking, let the support of a vector $\mathbf{x}$ of size N be the set of indices of nonzero entries: $\mathrm{supp}(\mathbf{x}) := \{j \in [N] : \mathbf{x}_j \neq 0\}$. We define $\mathrm{supp}(N, w) := \{\mathrm{supp}(\mathbf{x}) | \mathbf{x} \in \mathbb{F}_q^N, \mathsf{wt}(\mathbf{x}) = w\}$ as the set of the supports of all vectors of length N and weight w. The size of $\mathrm{supp}(N, w)$ is $\binom{N}{w}$. Out of all these possible supports, there are exactly $\binom{h-1}{\lceil \frac{w-i}{2} \rceil - 1}\binom{N-h}{\lfloor \frac{w-i}{2} \rfloor}\binom{h-\lceil \frac{w-i}{2} \rceil}{i}$ supports for each i that could lead to output weight h. This set of supports is denoted as $E_i \subset \mathrm{supp}(N, w)$. There are overlaps between E_is for different is as shown in [BFK+24], but it does not affect our analysis as an upperbound. For any support in E_i, there are exactly $(q-1)^{\lceil \frac{w-i}{2} \rceil}(q-2)^i$ possible values at the nonzero positions leading to $\mathsf{wt}(\mathbf{xA}) = h$. Moreover, these possible values are the same for all supports in E_i. See [AR11, BFK+24] for the computation of the IOWE with the property described here.

With this property, for any vector $\mathbf{x} \in \mathbb{F}_q^N$ of weight w, for a uniformly sampled permutation matrix $\mathbf{\Pi}$, the probability that $\mathrm{supp}(\mathbf{x\Pi}) \in E_i$ is $\frac{\binom{h-1}{\lceil \frac{w-i}{2} \rceil - 1}\binom{N-h}{\lfloor \frac{w-i}{2} \rfloor}\binom{h-\lceil \frac{w-i}{2} \rceil}{i}}{\binom{N}{w}}$. Conditioning on $\mathrm{supp}(\mathbf{x\Pi}) \in E_i$, for a uniformly sampled randomization matrix $\mathbf{V}$, the probability that $\mathsf{wt}(\mathbf{x\Pi VA}) = h$ is $\frac{(q-1)^{\lceil \frac{w-i}{2} \rceil}(q-2)^i}{q^w}$.

As a special case over the binary field $\mathbb{F}_2$, the IOWE becomes

$$A_{w,h}^{N,2} = \binom{h-1}{\lceil \frac{w}{2} \rceil - 1}\binom{N-h}{\lfloor \frac{w}{2} \rfloor}, \tag{2}$$

which is exactly the same as the first term of the sum in Equation (1) with $i = 0$ and $q = 2$.

Bounds of the binomial coefficients. We use 3 different variants of bounds for the binomial coefficients that appear in the IOWE. For low-weight messages, we use Stirling's approximation:

$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} \leq \left(\frac{en}{k}\right)^k.$$

For high-weight messages, we use a bound from [BCF+25]:

$$0.61664 \cdot 2^{nH(\frac{k}{n})} \leq \binom{n}{k} \leq 0.67352 \cdot \frac{2^{n \cdot H(\frac{k}{n})}}{\sqrt{n}},$$

where $H(x) = -x \log x - (1-x)\log(1-x)$ is the binary entropy function. We also use a bound from [KZKC08, AR09], which holds asymptotically as $N \to \infty$:

$$\left(\frac{n}{k}\right)^k \frac{1}{\varphi_n(k)} \leq \binom{n}{k} \leq \left(\frac{n}{k}\right)^k \varphi_k(k),$$

where $\varphi_n(k) = \exp\left(\frac{k(k-1)}{2n}\right)$.

3 Low-Weight Messages

In this section, we prove that for a randomly sampled RAA code over a large finite field, the probability that there exists a *low-weight* message with low codeword weight is very small. Combined with the high-weight case in Section 4, we prove that the RAA code is asymptotically good. Moreover, we show for the first time that the distance over a large finite field is much better than the RAA code over the binary field, with a much smaller failure probability. Formally speaking, we will prove the following theorem:

Theorem 2. *For a randomly sampled RAA code over $\mathbb{F}_q$ as defined in Section 2.2, for all nonzero messages $\mathbf{x}$ with weight $\mathsf{wt}(x) \leq \frac{n}{c' \log^2 n}$, for any constant relative distance $0 < \delta \leq \frac{1}{2}$,*

$$\Pr\left[\exists \mathbf{x} \ s.t. \ \mathsf{wt}(\mathbf{x} \cdot \mathbf{G}) \leq \delta N\right] \leq \tilde{O}(N^{-\eta}) + O(\frac{1}{q-1}), \tag{3}$$

where $\mathbf{G} = \mathbf{R} \cdot \mathbf{P}_1 \cdot \mathbf{A} \cdot \mathbf{P}_2 \cdot \mathbf{A}$ is the encoding matrix of the RAA code and the probability is over the random choices of $\mathbf{P}_1$ and $\mathbf{P}_2$.

That is, the output weights of all low-weight messages are larger than δN, except with a small probability that is inverse polynomial in N, plus a term that is inverse in q. The threshold $\frac{n}{c' \log^2 n}$ of low-weight messages will become clear in our proving techniques, and constants $c', \eta > 1$ will be specified later.

3.1 Fine-grained Analysis of the Accumulator Matrix

We start with the analysis of the *second* accumulator matrix over a large finite field. Suppose we have a linear error-correcting code that can be expressed as the product of some base matrix $\mathbf{M} \in \mathbb{F}_q^{n \times N}$, permutation matrix $\mathbf{P} = \mathbf{\Pi} \cdot \mathbf{V}$ and accumulator matrix $\mathbf{A}$. We would like to bound the probability that there exists a low-weight message $\mathbf{x}$ such that the weight of $\mathbf{x} \cdot \mathbf{M} \cdot \mathbf{P} \cdot \mathbf{A}$ is w_3, conditioning on the weight of $\mathbf{x} \cdot \mathbf{M}$ is w_2. We will show how to use this bound to prove our theorem in Section 3.2 and 3.3.

Lemma 1. *For a randomly sampled $\mathbf{P}$, for all nonzero messages $\mathbf{x}$ of weight w_0,*

$$\begin{aligned}&\Pr_{\mathbf{P}}[\exists \mathbf{x} \ s.t. \ \mathsf{wt}(\mathbf{xMPA}) = w_3 | \mathsf{wt}(\mathbf{x}) = w_0, \mathsf{wt}(\mathbf{xM}) = w_2] \\ &\leq \binom{n}{w_0} \sum_{i=w_2-2w_0-1}^{w_2-1} \frac{\binom{w_3-1}{\lceil \frac{w_2-i}{2} \rceil - 1}\binom{N-w_3}{\lfloor \frac{w_2-i}{2} \rfloor}\binom{w_3-\lceil \frac{w_2-i}{2} \rceil}{i}}{\binom{N}{w_2}} + O(\frac{1}{q-1}).\end{aligned}$$

Proof. By the union bound on the support (positions of nonzeros) of $\mathbf{x}$, we have

$$\begin{aligned}&\Pr_{\mathbf{P}}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{xMPA}) = w_3 | \mathsf{wt}(\mathbf{x}) = w_0, \mathsf{wt}(\mathbf{xM}) = w_2] \leq \\ &\sum_{s \in \text{supp}(n, w_0)} \Pr_{\mathbf{P}}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{xMPA}) = w_3 | \text{supp}(\mathbf{x}) = s, \mathsf{wt}(\mathbf{xM}) = w_2],\end{aligned}$$

where the sum is iterating through all possible supports with w_0 nonzeros in $\mathbf{x}$. We then add the condition of the positions of w_2 nonzeros in $\mathbf{xM\Pi}$ after the permutation matrix $\mathbf{\Pi}$. By the law of total probability, the equation above is equal to

$$\sum_{s\in\text{supp}(n,w_0)}\sum_{S\in\text{supp}(N,w_2)}\Pr_{\mathbf{P}}[\exists\mathbf{x}\text{ s.t. }\mathsf{wt}(\mathbf{xMPA})=w_3|\text{supp}(\mathbf{x})=s,$$
$$\text{supp}(\mathbf{xM\Pi})=S]\Pr_{\mathbf{\Pi}}[\text{supp}(\mathbf{xM\Pi})=S|\text{supp}(\mathbf{x})=s].$$

where S iterates through all possible supports with w_2 nonzeros in $\mathbf{xM\Pi}$. As $\mathbf{\Pi}$ is a uniformly sampled permutation, the probability of any input support s permuted to support S is $\frac{1}{\binom{N}{w_2}}$. Thus the equation above is equal to

$$=\frac{1}{\binom{N}{w_2}}\sum_{s\in\text{supp}(n,w_0)}\sum_{S\in\text{supp}(N,w_2)}\Pr_{\mathbf{P}}[\exists\mathbf{x}\text{ s.t. }\mathsf{wt}(\mathbf{xMPA})=w_3|\text{supp}(\mathbf{x})=s,$$
$$\text{supp}(\mathbf{xM\Pi})=S]$$
$$=\frac{\binom{n}{w_0}}{\binom{N}{w_2}}\sum_{S\in\text{supp}(N,w_2)}\Pr_{\mathbf{P}}[\exists\mathbf{x}\text{ s.t. }\mathsf{wt}(\mathbf{xMPA})=w_3|\text{supp}(\mathbf{xM\Pi})=S].$$

The second equality holds because $\mathsf{wt}(\mathbf{xMPA})=w_3|\text{supp}(\mathbf{xM\Pi})=S$ is independent of $\text{supp}(\mathbf{x})$ and w_0 now.

Recall that the IOWE of $\mathbf{A}$ with input weight w_2 and output weight w_3 is

$$A^{N,q}_{w_2,w_3}=\sum_{i=0}^{w_2-1}\binom{w_3-1}{\lceil\frac{w_2-i}{2}\rceil-1}\binom{N-w_3}{\lfloor\frac{w_2-i}{2}\rfloor}\binom{w_3-\lceil\frac{w_2-i}{2}\rceil}{i}(q-1)^{\lceil\frac{w_2-i}{2}\rceil}(q-2)^i,$$

where for each i, there are $\binom{w_3-1}{\lceil\frac{w_2-i}{2}\rceil-1}\binom{N-w_3}{\lfloor\frac{w_2-i}{2}\rfloor}\binom{w_3-\lceil\frac{w_2-i}{2}\rceil}{i}$ possible nonzero positions of the input vector that could lead to output weight w_3, denoted as E_i in Section 2.3. Other than these positions, the output weight cannot be w_3. Moreover, within each E_i, whether the output weight is w_3 or not is completely determined by the values of the input vector. Therefore, the equation above is

$$\leq\binom{n}{w_0}\sum_{i=0}^{w_2-1}\frac{\binom{w_3-1}{\lceil\frac{w_2-i}{2}\rceil-1}\binom{N-w_3}{\lfloor\frac{w_2-i}{2}\rfloor}\binom{w_3-\lceil\frac{w_2-i}{2}\rceil}{i}}{\binom{N}{w_2}}$$
$$\Pr_{\mathbf{V}}[\exists\mathbf{x}\text{ s.t. }\mathsf{wt}(\mathbf{xMPA})=w_3|\text{supp}(\mathbf{xM\Pi})\in E_i],$$

with the probability space over the sampling of $\mathbf{V}$ only.

We further divide this sum into two parts:

$$=\sum_{i=0}^{w_2-2w_0-2}\frac{\binom{n}{w_0}\binom{w_3-1}{\lceil\frac{w_2-i}{2}\rceil-1}\binom{N-w_3}{\lfloor\frac{w_2-i}{2}\rfloor}\binom{w_3-\lceil\frac{w_2-i}{2}\rceil}{i}}{\binom{N}{w_2}}$$
$$\Pr_{\mathbf{V}}[\exists\mathbf{x}\text{ s.t. }\mathsf{wt}(\mathbf{xMPA})=w_3|\text{supp}(\mathbf{xM\Pi})\in E_i],\qquad(4)$$

and

$$= \sum_{i=w_2-2w_0-1}^{w_2-1} \frac{\binom{n}{w_0}\binom{w_3-1}{\lceil\frac{w_2-i}{2}\rceil-1}\binom{N-w_3}{\lfloor\frac{w_2-i}{2}\rfloor}\binom{w_3-\lceil\frac{w_2-i}{2}\rceil}{i}}{\binom{N}{w_2}}$$
$$\Pr_{\mathbf{V}}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{xMPA}) = w_3 | \mathrm{supp}(\mathbf{xM\Pi}) \in E_i]. \tag{5}$$

In all of our cases, $w_2 > 2w_0$ and thus the two parts are well-defined. For Equation (4), we further apply the union bound on the possible values of $\mathbf{x}$. As $\mathbf{V}$ is uniformly sampled, for every value assignment of $\mathbf{x}$ with weight w_0, the nonzeros of $\mathbf{xM\Pi V}$ can be any w_2 values with the same probability of $\frac{1}{(q-1)^{w_2}}$. Therefore, by the union bound, $\Pr_{\mathbf{V}}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{xMPA}) = w_3 | \mathrm{supp}(\mathbf{xM\Pi}) \in E_i] \leq (q-1)^{w_0} \cdot \frac{(q-1)^{\lceil\frac{w_2-i}{2}\rceil}(q-2)^i}{(q-1)^{w_2}}$. When $i \leq w_2 - 2w_0 - 2$, the probability is $\leq \frac{1}{q-1}$, thus Equation (4) is $O(\frac{1}{q-1})$. This is why we divide the sum at $i = w_2 - 2w_0 - 2$.

For Equation (5), naively applying the union bound would lead to an equation that is larger than 1 for large q. Instead of the union bound, we simply apply the naive upperbound that any probability is no more than 1: $\Pr_{\mathbf{V}}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{xMPA}) = w_3 | \mathrm{supp}(\mathbf{xM\Pi}) \in E_i] \leq 1$. Intuitively, as long as the support of $\mathbf{xM\Pi}$ is in E_i after the permutation matrix $\mathbf{\Pi}$, we count it as having output weight w_3 regardless of the values of $\mathbf{x}$ and $\mathbf{xM\Pi V}$, which is clearly an upperbound of the probability. This technique was proposed in [BFK+24], but was applied to the entire range of the summation instead of Equation (5). We will see later that our fine-grained analysis is critical to obtain the result of our paper.

Combining the two parts, the probability

$$\Pr_{\mathbf{P}}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{xMPA}) = w_3 | \mathsf{wt}(\mathbf{x}) = w_0, \mathsf{wt}(\mathbf{xM}) = w_2]$$
$$\leq \binom{n}{w_0} \sum_{i=w_2-2w_0-1}^{w_2-1} \frac{\binom{w_3-1}{\lceil\frac{w_2-i}{2}\rceil-1}\binom{N-w_3}{\lfloor\frac{w_2-i}{2}\rfloor}\binom{w_3-\lceil\frac{w_2-i}{2}\rceil}{i}}{\binom{N}{w_2}} + O(\frac{1}{q-1}),$$

which completes the proof. □

Treatment of the term $O(\frac{1}{q-1})$. As shown in the proof above, the term $O(\frac{1}{q-1})$ hides factors that depend on N. Technically, the term could still be large even if $q >> N$. However, as shown in [BFK+24], these factors related to N are also inverse polynomials in N for some parameter regimes. Moreover, as i goes to 0, the term decays exponentially in $q-1$. Therefore, we view the term as extremely small and focus on the combinatorial terms in the rest of this paper. We believe with careful analysis, it is possible to show that this term is indeed negligible as long as $q = 2^{\omega(\log N)}$ as in cryptographic applications. In addition, we demonstrate empirically in Section 5 that this term is negligible via scripts calculating the exact failure probability.

Before completing this section, we introduce two more lemmas for the upperbound above. For ease of notation, we denote the numerator of the combinatorial terms in the summand as

$$S(i, w_2, w_3) := \binom{w_3 - 1}{\lceil \frac{w_2-i}{2} \rceil - 1} \binom{N - w_3}{\lfloor \frac{w_2-i}{2} \rfloor} \binom{w_3 - \lceil \frac{w_2-i}{2} \rceil}{i}$$

Note that the other two combinatorial terms are independent of i and w_3. Also recall that in this section, we consider the input weight $w_0 \in [1, cn/\log^2 n]$.

Lemma 2. *For all $w_3 \leq N/2$, $S(i, w_2, w_3)$ is increasing with w_3.*

Proof. We will consider the ratio between consecutive terms over w_3 and show that the ratio is strictly greater than 1 under our conditions.

$$\begin{aligned}
\frac{S(i, w_2, w_3 + 1)}{S(i, w_2, w_3)} &= \frac{\binom{w_3}{\lceil \frac{w_2-i}{2} \rceil - 1} \binom{N - w_3 - 1}{\lfloor \frac{w_2-i}{2} \rfloor} \binom{w_3 - \lceil \frac{w_2-i}{2} \rceil + 1}{i}}{\binom{w_3 - 1}{\lceil \frac{w_2-i}{2} \rceil - 1} \binom{N - w_3}{\lfloor \frac{w_2-i}{2} \rfloor} \binom{w_3 - \lceil \frac{w_2-i}{2} \rceil}{i}} \\
&= \frac{w_3}{w_3 - \lceil \frac{w_2-i}{2} \rceil + 1} \frac{N - w_3 - \lfloor \frac{w_2-i}{2} \rfloor}{N - w_3} \frac{w_3 - \lceil \frac{w_2-i}{2} \rceil + 1}{w_3 - \lceil \frac{w_2-i}{2} \rceil - i + 1} \\
&= \frac{w_3}{w_3 - \lceil \frac{w_2-i}{2} \rceil - i + 1} \frac{N - w_3 - \lfloor \frac{w_2-i}{2} \rfloor}{N - w_3}
\end{aligned}$$

To show that this is increasing with w_3, we need to have the following:

$$\begin{aligned}
&\frac{w_3}{w_3 - \lceil \frac{w_2-i}{2} \rceil - i + 1} \frac{N - w_3 - \lfloor \frac{w_2-i}{2} \rfloor}{N - w_3} > 1 \\
&\Leftrightarrow w_3 \left(N - w_3 - \lfloor \frac{w_2 - i}{2} \rfloor \right) > (N - w_3) \left(w_3 - \lceil \frac{w_2 - i}{2} \rceil - i + 1 \right) \\
&\Leftrightarrow w_3 \left(N - w_3 \right) - w_3 \left(\lfloor \frac{w_2 - i}{2} \rfloor \right) > w_3(N - w_3) - (N - w_3) \left(\lceil \frac{w_2 - i}{2} \rceil + i - 1 \right) \\
&\Leftrightarrow w_3 \left(\lfloor \frac{w_2 - i}{2} \rfloor \right) < (N - w_3) \left(\lceil \frac{w_2 - i}{2} \rceil + i - 1 \right)
\end{aligned}$$

Since we are considering $i \in [w_2 - 2w_0 - 1, w_2 - 1]$, a sufficient condition can be derived by maximizing the LHS over i and minimizing the RHS:

$$\begin{aligned}
w_0 w_3 &< (N - w_3)(w_2 - 2w_0 - 1) \\
w_3 &< N \frac{w_2 - 2w_0 - 1}{w_2 - w_0 - 1}
\end{aligned}$$

Thus, we can say that $S(i, w_2, w_3)$ is increasing with w_3 for $w_3 \leq N/2$. $\square$

Lemma 3. *Let $w_2 > 4w_0$. Then, the summand $S(i, w_2, \delta N)$ is decreasing with i on the interval $i \in [w_2 - 2w_0 - 1, w_2 - 1]$.*

Proof. We will split this into cases corresponding to the parity of $w_2 - i$.
Case 1 (odd $w_2 - i$). Let $w_2 - i = 2m + 1$, where m is an integer in the interval $[0, w_0]$. We will consider the ratio between consecutive terms over i and show that it is strictly less than 1. We can use the following two properties of the binomial coefficient: $\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}$, $\binom{n-1}{k} = \frac{n-k}{n}\binom{n}{k}$.

$$\frac{S(i+1, w_2, \delta N)}{S(i, w_2, \delta N)} = \frac{\binom{\delta N-1}{m-1}\binom{N-\delta N}{m}\binom{\delta N-m}{i+1}}{\binom{\delta N-1}{m}\binom{N-\delta N}{m}\binom{\delta N-m-1}{i}} = \frac{m}{\delta N - m}\frac{\delta N - m}{i+1} = \frac{m}{w_2 - 2m}$$

In order to have $\frac{m}{w_2-2m} < 1$, we need $w_2 > 3m$, or $w_2 > 3w_0$.

Case 2 (even $w_2 - i$). Let $w_2 - i = 2m$, where m is now in the interval $[1, w_0]$. We will again compute the ratio between consecutive terms over i:

$$\begin{aligned}
\frac{S(i+1, w_2, \delta N)}{S(i, w_2, \delta N)} &= \frac{\binom{\delta N-1}{m-1}\binom{N-\delta N}{m-1}\binom{\delta N-m}{i+1}}{\binom{\delta N-1}{m-1}\binom{N-\delta N}{m}\binom{\delta N-m}{i}} \\
&= \frac{m}{N - \delta N - m + 1}\frac{\delta N - m - i}{i+1} \\
&\leq \frac{\delta N}{(1-\delta)N - w_0 + 1}\frac{w_0}{w_2 - 2w_0 + 1} \\
&\leq \frac{\delta N}{(1-\delta)N/2}\frac{w_0}{w_2 - 2w_0} \\
&= \frac{2\delta}{1-\delta}\frac{w_0}{w_2 - 2w_0},
\end{aligned}$$

where the second to last line follows from $w_0 \leq N/4$, which is the case since $r \geq 4$. Note that $\frac{2\delta}{1-\delta}$ is at most 2 since $\delta \leq 1/2$. If we assume that $w_2 > 4w_0$, then we can say that $w_2 - 2w_0 > 2w_0$, and we can conclude the following:

$$\frac{2\delta}{1-\delta}\frac{w_0}{w_2 - 2w_0} < 2 \cdot \frac{w_0}{2w_0} = 1.$$

Thus the ratio between consecutive terms in both cases is strictly less than 1, so $S(i, w_2, \delta N)$ is decreasing with i. □

3.2 Output Weight of Accumulation with Amplification in Stage 1

With the result of Section 3.1, we divide the analysis of Theorem 2 into two stages. The first stage is the RA code part, which corresponds to multiplying a message $\mathbf{x}$ by $\mathbf{M} = \mathbf{R}\mathbf{P}_1\mathbf{A}$. The second stage is further multiplying the result of stage 1 by $\mathbf{P}_2\mathbf{A}$. We consider stage 2 in this section with the assumption that stage 1 amplifies the weight of message from w_0 to $w_2 > cw_0 \log N$ for all low weight messages. As mentioned in the introduction, this amplification is inspired by the analysis in [BFK+24] from the property of an expander matrix in the EA code. However, the encoding time of the EA code is $O(n \log n)$ and the analysis only works for low-weight messages. We will show in Section 3.3 that one can get a similar amplification using the stage 1 of an RAA code, and further show in Section 4 that the distance is also good for the high-weight case.

Lemma 4. *For a randomly sampled $\mathbf{P}_2$, for all messages $\mathbf{x}$ with weight $0 < w_0 \leq \frac{n}{c' \log^2 n}$, suppose the weight $\mathsf{wt}(\mathbf{xM}) = w_2 > cw_0 \log N$, then for any constant relative distance $0 < \delta \leq \frac{1}{2}$*

$$\Pr_{\mathbf{P}_2}[\exists \mathbf{x} \textit{ s.t. } \mathsf{wt}(\mathbf{xMP}_2\mathbf{A}) \leq \delta N] \leq O(N^{-\eta_1}) + O(\frac{1}{q-1}).$$

Proof. By Lemma 1, and the union bound on w_0, w_2, w_3,

$$\begin{aligned}
&\Pr_{\mathbf{P}_2}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{xMP}_2\mathbf{A}) \leq \delta N] \\
\leq &\sum_{w_0=1}^{n/c'\log^2 n} \sum_{w_2=cw_0\log N+1}^{N} \sum_{w_3=1}^{\delta N} \Pr_{\mathbf{P}_2}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{xMP}_2\mathbf{A}) = w_3 | \mathsf{wt}(\mathbf{x}) = w_0, \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \mathsf{wt}(\mathbf{xM}) = w_2] \\
\leq &\sum_{w_0=1}^{n/c'\log^2 n} \sum_{w_2=cw_0\log N+1}^{N} \sum_{w_3=1}^{\delta N} \\
&\qquad \left[\binom{n}{w_0} \sum_{i=w_2-2w_0-1}^{w_2-1} \frac{\binom{w_3-1}{\lceil\frac{w_2-i}{2}\rceil-1}\binom{N-w_3}{\lfloor\frac{w_2-i}{2}\rfloor}\binom{w_3-\lceil\frac{w_2-i}{2}\rceil}{i}}{\binom{N}{w_2}} + O(\frac{1}{q-1})\right]
\end{aligned}$$

$$\begin{aligned}
= &\sum_{w_0=1}^{n/c'\log^2 n} \sum_{w_2=cw_0\log N+1}^{N} \sum_{w_3=1}^{\delta N} \\
&\left[\binom{n}{w_0} \sum_{i=w_2-2w_0-1}^{w_2-1} \frac{\binom{w_3-1}{\lceil\frac{w_2-i}{2}\rceil-1}\binom{N-w_3}{\lfloor\frac{w_2-i}{2}\rfloor}\binom{w_3-\lceil\frac{w_2-i}{2}\rceil}{i}}{\binom{N}{w_2}}\right] + O(\frac{1}{q-1}).
\end{aligned} \tag{6}$$

We focus on the combinatorial terms. As discussed in Section 2.3, in order for the numerator to be larger than 0, $i \leq 2w_3 - w_2 + 1$, while $i \geq w_2 - 2w_0 - 1$ in the summation. Therefore, the summand is 0 for $w_2 > w_3 + w_0 + 1$. Again, w_2 can be as big as $2w_3$, but those cases happen for $i < w_2 - 2w_0 - 1$ and are in the $O(\frac{1}{q-1})$ terms.

As $\delta \leq \frac{1}{2}$, by Lemma 2 and Lemma 3, we take $w_3 = \delta N$ and $i = w_2 - 2w_0 - 1$ as the maximum and we have the following:

$$
\begin{aligned}
&\sum_{w_3=1}^{\delta N}\binom{n}{w_0}\sum_{i=w_2-2w_0-1}^{w_2-1}\frac{\binom{w_3-1}{\lceil\frac{w_2-i}{2}\rceil-1}\binom{N-w_3}{\lfloor\frac{w_2-i}{2}\rfloor}\binom{w_3-\lceil\frac{w_2-i}{2}\rceil}{i}}{\binom{N}{w_2}} \\
&\le \delta N(2w_0+1)\binom{n}{w_0}\frac{\binom{\delta N-1}{w_0}\binom{N-\delta N}{w_0}\binom{\delta N-w_0-1}{w_2-2w_0-1}}{\binom{N}{w_2}} \\
&\le (2w_0+1)(w_2-2w_0)\binom{n}{w_0}\frac{\binom{\delta N}{w_0}\binom{N-\delta N}{w_0}\binom{\delta N-w_0}{w_2-2w_0}}{\binom{N}{w_2}} \qquad (7)
\end{aligned}
$$

We have the two cases below:

Case 1: $w_2 \le \delta N/2$. In this case, it is easy to show that $\frac{\binom{\delta N-w_0}{w_2-2w_0}}{\binom{N}{w_2}} \le \delta^{w_2}$. First, notice that $\binom{\delta N-w_0}{w_2-2w_0} \le \binom{\delta N}{w_2}$ for $w_2 \le \delta N/2$. It follows that

$$
\frac{\binom{\delta N-w_0}{w_2-2w_0}}{\binom{N}{w_2}} \le \frac{\binom{\delta N}{w_2}}{\binom{N}{w_2}} = \prod_{i=0}^{w_2-1}\frac{\delta-i/N}{1-i/N} \le \delta^{w_2}
$$

Using this and Stirling's Approximation $\left(\frac{N}{w}\right)^w \le \binom{N}{w} \le \left(\frac{eN}{w}\right)^w$, we can upper-bound Equation (7) by the following:

$$
\le (2w_0+1)(w_2-2w_0)\left(\frac{eN}{rw_0}\right)^{w_0}\left(\frac{e\delta N}{w_0}\right)^{w_0}\left(\frac{e(1-\delta)N}{w_0}\right)^{w_0}\delta^{w_2}.
$$

As $w_2 > cw_0 \log n$, the equation above is

$$
\begin{aligned}
&\le (2w_0+1)(cw_0\log N - 2w_0)\left(\frac{e^3}{rw_0^3}\right)^{w_0}(1-\delta)^{w_0}\delta^{cw_0\log N}N^{3w_0} \\
&\le \frac{ce^{3w_0}(2+1/w_0)}{w_0^{3w_0-2}}(1-\delta)^{w_0}N^{w_0(3-c\log(1/\delta))}\log N. \qquad (8)
\end{aligned}
$$

For example, by setting $c \ge \frac{6}{-\log\delta}$, the probability above is $\le N^{-3w_0}\log N$. Summing over $w_2 \in [cw_0\log N+1, \delta N/2]$ introduces at most a factor of N, thus the probability is still $\le N^{1-3w_0}\log N$.

Finally, summing over all $w_0 \in [1, n/c'\log^2 n]$, the total probability is upper-bounded by $\tilde{O}(N^{-\eta_1})$ for $\eta_1 = -c\log\delta - 4$, where the dominating term is by $w_0 = 1$.

Case 2: $w_2 > \delta N/2$. As explained above, $w_2 \le w_3 + w_0 + 1 = \delta N + w_0 + 1$. In this case, as $\delta \le \frac{1}{2}$ and $w_0 \le \frac{n}{c'\log^2 n}$, $\frac{\binom{\delta N-w_0}{w_2-2w_0}}{\binom{N}{w_2}} \le \frac{\binom{\delta N-w_0}{(\delta N-w_0)/2}}{\binom{N}{\delta N/2}} \le \frac{\binom{\delta N}{\delta N/2}}{\binom{N}{\delta N/2}}$, where the numerator is taking the maximum of the combinatorial term at $\binom{\delta N}{\delta N/2}$, and the denominator is taking the minimum at $w_2 = \delta N/2$. By the same argument above, the ratio is $\le \delta^{\delta N/2}$.

Using this and Stirling's Approximation, Equation (7) is upper-bounded by:

$$\begin{aligned}
&\leq (2w_0+1)(w_2-2w_0)\left(\frac{eN}{rw_0}\right)^{w_0}\left(\frac{e\delta N}{w_0}\right)^{w_0}\left(\frac{e(1-\delta)N}{w_0}\right)^{w_0}\delta^{\delta N/2}. \\
&\leq (2w_0+1)(\delta N+1-w_0)\left(\frac{e^3}{rw_0^3}\right)^{w_0}(1-\delta)^{w_0}\delta^{\delta N/2}N^{3w_0} \\
&\leq (2w_0+1)(\delta N+1)\frac{e^{3w_0}}{(rw_0^3)^{w_0}}(1-\delta)^{w_0}N^{3w_0-\frac{\delta N\log(1/\delta)}{2\log N}}.
\end{aligned}$$

As $w_0 \leq \frac{n}{c'\log^2 n}$, the probability is $O(N^{\frac{\delta N\log(1/\delta)}{2\log N}}) = O(\delta^{\delta N/2})$, which is negligible. This completes the proof together with Case 1. □

In Appendix A.1, we show that the upperbound of the probability in Lemma 4 over a large finite field $\mathbb{F}_q$ is strictly smaller than the provable bound over the binary field $\mathbb{F}_2$.

3.3 Probability of Amplification in Stage 1

In this section, we further prove that with high probability, the weights of all low-weight messages are amplified by the first stage of the RAA code.

Lemma 5. *For a randomly sampled* $\mathbf{P}_1$*, for all messages* $\mathbf{x}$ *with weight* $0 < w_0 \leq \frac{n}{c'\log^2 n}$,

$$\Pr_{\mathbf{P}_1}[\exists \mathbf{x} \ s.t.\ \mathsf{wt}(\mathbf{x}\mathbf{R}\mathbf{P}_1\mathbf{A}) \leq cw_0\log N] \leq \tilde{O}(N^{-\eta_2}) + O(\frac{1}{q-1}).$$

Proof. We first invoke Lemma 1 and the union bound. Note that though Lemma 1 is stated for the second accumulator matrix, the same bound holds for the first accumulator matrix with $\mathbf{M} = \mathbf{R}$ and (w_2, w_3) replaced by (w_1, w_2). No special properties of (w_2, w_3) are used in the proof that are not satisfied by (w_1, w_2). Therefore:

$$\begin{aligned}
&\Pr_{\mathbf{P}_1}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{x}\mathbf{R}\mathbf{P}_1\mathbf{A}) \leq cw_0\log N] \\
&\leq \sum_{w_0=1}^{n/c'\log^2 n}\sum_{w_2=1}^{cw_0\log N} \\
&\qquad \left[\binom{n}{w_0}\sum_{i=w_1-2w_0-1}^{w_1-1}\frac{\binom{w_2-1}{\lceil\frac{w_1-i}{2}\rceil-1}\binom{N-w_2}{\lfloor\frac{w_1-i}{2}\rfloor}\binom{w_2-\lceil\frac{w_1-i}{2}\rceil}{i}}{\binom{N}{w_1}} + O(\frac{1}{q-1})\right] \\
&= \sum_{w_0=1}^{n/c'\log^2 n}\sum_{w_2=1}^{cw_0\log N}\left[\binom{n}{w_0}\sum_{i=w_1-2w_0-1}^{w_1-1}\frac{\binom{w_2-1}{\lceil\frac{w_1-i}{2}\rceil-1}\binom{N-w_2}{\lfloor\frac{w_1-i}{2}\rfloor}\binom{w_2-\lceil\frac{w_1-i}{2}\rceil}{i}}{\binom{N}{w_1}}\right] \\
&\qquad + O(\frac{1}{q-1}). \qquad (9)
\end{aligned}$$

where $w_1 = rw_0 \geq 4w_0$. We again focus on the combinatorial terms:

$$\sum_{w_0=1}^{n/c'\log^2 n} \sum_{w_2=1}^{cw_0\log N} \binom{n}{w_0} \sum_{i=(r-2)w_0-1}^{rw_0-1} \frac{\binom{w_2-1}{\lceil\frac{rw_0-i}{2}\rceil-1}\binom{N-w_2}{\lfloor\frac{rw_0-i}{2}\rfloor}\binom{w_2-\lceil\frac{rw_0-i}{2}\rceil}{i}}{\binom{N}{rw_0}}. \tag{10}$$

By Lemma 2, as $w_2 \leq N/2$, the numerator of the summand $S(i, w_1, w_2)$ increases with w_2, and thus we bound Equation (10) by the maximum of w_2 at $cw_0 \log N$:

$$\sum_{w_0=1}^{n/c'\log^2 n} (cw_0\log N)\binom{n}{w_0} \sum_{i=(r-2)w_0-1}^{rw_0-1} \frac{\binom{cw_0\log N-1}{\lceil\frac{rw_0-i}{2}\rceil-1}\binom{N-cw_0\log N}{\lfloor\frac{rw_0-i}{2}\rfloor}\binom{cw_0\log N-\lceil\frac{rw_0-i}{2}\rceil}{i}}{\binom{N}{rw_0}}$$

Similar to the argument in Lemma 3, the numerator decreases with i as well. The formal proof can be found in Lemma 8 in the appendix. Therefore, we can bound the equation above as

$$\leq \sum_{w_0=1}^{n/c'\log^2 n} (2w_0+1)(cw_0\log N)\binom{n}{w_0} \frac{\binom{cw_0\log N-1}{w_0}\binom{N-cw_0\log N}{w_0}\binom{cw_0\log N-w_0}{(r-2)w_0-1}}{\binom{N}{rw_0}}$$

Consider the individual summand for a fixed w_0. By Stirling's inequality, together with the simple bound of $(2w_0 + 1) \leq 3w_0$ and the equality of the binomial coefficient: $\binom{n-1}{k} = \frac{n-k}{n}\binom{n}{k}$, $\binom{n}{k} = \frac{n-k+1}{k}\binom{n}{k-1}$, we have

$$\begin{aligned}
&(2w_0+1)(cw_0\log N)\binom{n}{w_0} \frac{\binom{cw_0\log N-1}{w_0}\binom{N-cw_0\log N}{w_0}\binom{cw_0\log N-w_0}{(r-2)w_0-1}}{\binom{N}{rw_0}} \\
&= (2w_0+1)(cw_0\log N - w_0)\frac{(r-2)w_0}{cw_0\log N-(r-1)w_0+1} \\
&\qquad \cdot \binom{n}{w_0}\frac{\binom{cw_0\log N}{w_0}\binom{N-cw_0\log N}{w_0}\binom{cw_0\log N-w_0}{(r-2)w_0}}{\binom{N}{rw_0}} \\
&\leq \frac{3w_0^2(r-2)}{1-(r-1)/c\log N}\left(\frac{en}{w_0}\right)^{w_0}\left(\frac{w_0}{n}\right)^{rw_0}\left(\frac{cew_0\log N}{w_0}\right)^{w_0} \\
&\qquad \cdot \left(\frac{e(N-cw_0\log N)}{w_0}\right)^{w_0}\left(\frac{e(cw_0\log N-w_0)}{(r-2)w_0}\right)^{(r-2)w_0} \\
&\leq \frac{3w_0^2(r-2)}{1-(r-1)/c\log N}\left(\frac{eN}{rw_0}\right)^{w_0}\left(\frac{rw_0}{N}\right)^{rw_0}(ce\log N)^{w_0} \\
&\qquad \cdot \left(\frac{eN}{w_0}\right)^{w_0}\left(\frac{ec\log N}{r-2}\right)^{(r-2)w_0} \\
&\leq \frac{3w_0^2(r-2)}{1-(r-1)/c\log N}\left[\frac{e^{r+1}w_0^{r-2}r^{r-1}}{(r-2)^{r-2}}\frac{(c\log N)^{r-1}}{N^{r-2}}\right]^{w_0}
\end{aligned}$$

$$= \frac{3w_0^2(r-2)}{1-(r-1)/c\log N}\left(\frac{e^{r+1}r^{r-1}}{(r-2)^{r-2}}\right)^{w_0}\left(\frac{w_0^{r-2}(c\log N)^{r-1}}{N^{r-2}}\right)^{w_0}. \quad (11)$$

Since e, c, r are constants, the equation above is equal to

$$O\left(w_0^2(g\log N)^{w_0}(\frac{w_0 c\log N}{N})^{(r-2)w_0}\right), \quad (12)$$

where $g = \frac{e^{r+1}r^{r-1}}{(r-2)^{r-2}}$ is a constant.

We further differentiate the following two cases:

Case 1: w_0 is from 1 to $O(\log N)$. In this case, the numerator is $\mathsf{polylog}(N)^{w_0}$ while the denominator is $\mathsf{poly}(N)^{w_0}$. Therefore, the upperbound is $O\left(\frac{(\log N)^{(r-1)}}{N^{(r-2)}}\right)$ after summing over all w_0, and the dominating terms is $w_0 = 1$. This is consistent with the intuition that the nonzero messages with the least weight of 1 is the most challenging to amplify.

Case 2: $w_0 = \omega(\log N)$. In this case, for any $w_0 \le n/c'\log^2 n$, as $N = rn$, Equation (12) is $\le O\left(w_0^2(\frac{c^{r-2}g}{(rc')^{r-2}})^{w_0}\frac{1}{\log N}^{(r-3)w_0}\right)$. As $r \ge 4$, c, r, c', g are all constants, it is $O\left((\frac{c''}{(\log N)^{(r-3)}})^{w_0}\right)$ for $c'' = \frac{c^{r-2}g}{(rc')^{r-2}}$, which is negligible for $w_0 = \omega(\log N)$. As Equation (12) is summed over $w_0 = \omega(\log N)$ to $\frac{n}{c'\log^2 n}$, the probability is still negligible after the summation.

Combining the two cases completes the proof of Lemma 5, where $\eta_2 = r-2$ due to the first case dominated by $w_0 = 1$. □

Moreover, Equation (12) explains the reason of the threshold $\frac{n}{c'\log^2 n}$ of low-weight messages. The term $\frac{w_0 c\log N}{N}$ needs to be at most $\frac{1}{\log N}$ to cancel other terms in the numerator and make the bound smaller than 1.

In Appendix A.2, we prove a bound for the amplification of the first stage of the RAA code over the binary field where the failure probability is $\tilde{O}(N^{1-\frac{r}{2}})$. As explained in the introduction, this is a better result for the binary RAA than the bound in [BCF+25, Theorem 9.1], where the failure probability is $O(N^{1+\gamma-\frac{r}{2}})$ for a constant $0 < \gamma < 1$. In either case, we clearly show that the failure probability over a large finite field ($\tilde{O}(N^{2-r})$) is strictly smaller than the RAA code over the binary field. For example, when $r = 4$, the failure probability over a large finite field is $\tilde{O}(N^{-2})$, which is already good enough for large N, while the failure probability over the binary field is only $\tilde{O}(N^{-1})$.

Finally, by combining Lemma 4 in Section 3.2 and Lemma 5 in Section 3.3 through the law of total probability, we are able to prove Theorem 2 as:

$$\begin{aligned}
&\Pr\left[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{x}\cdot\mathbf{G}) \le \delta N\right]\\
=&\Pr\left[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{x}\cdot\mathbf{G}) \le \delta N \middle| \text{ Stage 1 succeeds }\right]\cdot\Pr[\text{ Stage 1 succeeds }]\\
&+\Pr\left[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{x}\cdot\mathbf{G}) \le \delta N \middle| \text{ Stage 1 fails }\right]\cdot\Pr[\text{ Stage 1 fails }]\\
\le&\Pr\left[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{x}\cdot\mathbf{G}) \le \delta N \middle| \text{ Stage 1 succeeds }\right]+\Pr[\text{ Stage 1 fails }]\\
=&\tilde{O}(N^{-\eta_1})+\tilde{O}(N^{-\eta_2})+O(\frac{1}{q-1}) = \tilde{O}(N^{-\eta})+O(\frac{1}{q-1}),
\end{aligned}$$

for $\eta = \min(\eta_1, \eta_2)$.

4 High-Weight Messages

In this section, we prove that for a randomly sampled RAA code over a large finite field, the probability that there exists a *high-weight* message with low codeword weight is negligible, which completes the full distance analysis of the RAA code.

Theorem 3. *For a randomly sampled RAA code over $\mathbb{F}_q$ as defined in Section 2.2, for all nonzero messages $\mathbf{x}$ with weight $\mathsf{wt}(x) > \frac{n}{c' \log^2 n}$, for any constant relative distance $0 < \delta \leq \frac{1}{2}$,*

$$\Pr\left[\exists \mathbf{x} \text{ } s.t. \text{ } \mathsf{wt}(\mathbf{x} \cdot \mathbf{G}) \leq \delta N\right] \leq \exp(-O(N/\log N)) + O(\frac{1}{q-1}),$$

where $\mathbf{G} = \mathbf{R} \cdot \mathbf{P}_1 \cdot \mathbf{A} \cdot \mathbf{P}_2 \cdot \mathbf{A}$ is the encoding matrix of the RAA code and the probability is over the random choices of $\mathbf{P}_1$ and $\mathbf{P}_2$.

To prove this theorem, by the union bound and the law of total probability, the probability is upperbounded by:

$$\begin{aligned}
&\Pr\left[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{x} \cdot \mathbf{G}) \leq \delta N\right] \\
&\leq \sum_{w_0 = \frac{n}{c' \log^2 n}+1}^{n} \binom{n}{w_0} \Pr_{\mathbf{P}_1, \mathbf{P}_2}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{x}\mathbf{G} \leq \delta N | \mathsf{wt}(\mathbf{x}) = w_0] \\
&= \sum_{w_0 = \frac{n}{c' \log^2 n}+1}^{n} \binom{n}{w_0} \sum_{w_2 = \lfloor \frac{r w_0}{2} \rfloor}^{2\delta N} \Pr_{\mathbf{P}_1}[\mathsf{wt}(\mathbf{x}\mathbf{R}\mathbf{P}_1\mathbf{A}) = w_2 | \mathsf{wt}(\mathbf{x}) = w_0] \\
&\qquad \cdot \Pr_{\mathbf{P}_2}[\mathsf{wt}(\mathbf{x}\mathbf{G}) \leq \delta N | \mathsf{wt}(\mathbf{x}) = w_0, \mathsf{wt}(\mathbf{x}\mathbf{R}\mathbf{P}_1\mathbf{A}) = w_2] \\
&\leq \sum_{w_0 = \frac{n}{c' \log^2 n}+1}^{n} \binom{n}{w_0} \sum_{w_2 = \lfloor \frac{r w_0}{2} \rfloor}^{2\delta N} \Pr_{\mathbf{P}_1}[\mathsf{wt}(\mathbf{x}\mathbf{R}\mathbf{P}_1\mathbf{A}) = w_2 | \mathsf{wt}(\mathbf{x}) = w_0] \\
&\qquad \cdot \sum_{w_3 = \lfloor \frac{w_2}{2} \rfloor}^{\delta N} \Pr_{\mathbf{P}_2}[\mathsf{wt}(\mathbf{x}\mathbf{G}) = w_3 | \mathsf{wt}(\mathbf{x}) = w_0, \mathsf{wt}(\mathbf{x}\mathbf{R}\mathbf{P}_1\mathbf{A}) = w_2]
\end{aligned}$$

The ranges of w_2 and w_3 are due to the property of the accumulator matrix $\mathbf{A}$ where the output weight cannot be smaller than half of the input weight. Recall that $w_1 = r w_0$ after the repetition matrix.

Then by Lemma 1 applied to both probabilities, the equation above is

$$
\begin{aligned}
\leq \sum_{w_0=\frac{n}{c'\log^2 n}+1}^{n} \sum_{w_2=\lfloor\frac{rw_0}{2}\rfloor}^{2\delta N} \sum_{w_3=\lfloor\frac{w_2}{2}\rfloor}^{\delta N} \binom{n}{w_0} \\
\cdot \Big(\sum_{i=w_1-2w_0-1}^{w_1-1} \frac{\binom{N-w_2}{\lceil\frac{w_1-i}{2}\rceil}\binom{w_2-1}{\lfloor\frac{w_1-i}{2}\rfloor}\binom{w_2-\lceil\frac{w_1-i}{2}\rceil}{i}}{\binom{N}{w_1}} + O(\frac{1}{q-1})\Big) \\
\cdot \Big(\sum_{j=w_2-2w_0-1}^{w_2-1} \frac{\binom{N-w_3}{\lceil\frac{w_2-j}{2}\rceil}\binom{w_3-1}{\lfloor\frac{w_2-j}{2}\rfloor}\binom{w_3-\lceil\frac{w_2-j}{2}\rceil}{j}}{\binom{N}{w_2}} + O(\frac{1}{q-1})\Big)
\end{aligned} \tag{13}
$$

Again focusing on the combinatorial terms, the probability we are trying to bound is

$$
\begin{aligned}
\sum_{w_0=\frac{n}{c'\log^2 n}+1}^{n} \sum_{w_2=\lfloor\frac{rw_0}{2}\rfloor}^{2\delta N} \sum_{w_3=\lfloor\frac{w_2}{2}\rfloor}^{\delta N} \sum_{i=w_1-2w_0-1}^{w_1-1} \sum_{j=w_2-2w_0-1}^{w_2-1} \binom{n}{w_0} \\
\frac{\binom{N-w_2}{\lceil\frac{w_1-i}{2}\rceil}\binom{w_2-1}{\lfloor\frac{w_1-i}{2}\rfloor}\binom{w_2-\lceil\frac{w_1-i}{2}\rceil}{i}}{\binom{N}{w_1}} \cdot \frac{\binom{N-w_3}{\lceil\frac{w_2-j}{2}\rceil}\binom{w_3-1}{\lfloor\frac{w_2-j}{2}\rfloor}\binom{w_3-\lceil\frac{w_2-j}{2}\rceil}{j}}{\binom{N}{w_2}}.
\end{aligned} \tag{14}
$$

Note that the summand is exactly the same as the bound in [AHKM25, Appendix A]. However, the ranges of w_0, w_2, w_3 are different due to our fine-grained analysis, which are crucial to prove that the bound is negligible for high-weight messages in our case.

Using the variant of the IOWE in [AR11], by a change of variables on i, j, the equation above is equivalent to

$$
\begin{aligned}
\sum_{w_0=\frac{n}{c'\log^2 n}+1}^{n} \sum_{w_2=\lfloor\frac{rw_0}{2}\rfloor}^{2\delta N} \sum_{w_3=\lfloor\frac{w_2}{2}\rfloor}^{\delta N} \sum_{i=0}^{w_0}\sum_{j=0}^{w_0} \binom{n}{w_0} \\
\frac{\binom{N-w_2}{i}\binom{w_2-1}{i}\binom{w_2-i}{w_1-2i}}{\binom{N}{w_1}} \cdot \frac{\binom{N-w_3}{j}\binom{w_3-1}{j}\binom{w_3-j}{w_2-2j}}{\binom{N}{w_2}},
\end{aligned} \tag{15}
$$

which will be the equation we bound in the rest of this section. Following the approach in [AR09], we write w_0, w_2, w_3, i, j in terms of N as:

$$
w_0 = \alpha N^a \qquad w_2 = \beta N^b \qquad w_3 = \rho N^c \qquad i = \gamma_1 N^{e_1} \qquad j = \gamma_2 N^{e_2}.
$$

With these, $w_1 = rw_0 = r\alpha N^a$. Moreover, as $w_0 > \frac{n}{c'\log n}$ and the output weight of the accumulator matrix is no less than half of its input weight, we have $1 - \frac{2\log\log N}{\log N} \leq a \leq b \leq c \leq 1$. $e_1 \leq a$ and $e_2 \leq a$ due to the ranges of i and j in Equation (15).

To prove that Equation (15) is negligible, we follow the approach in [AR09] and differentiate between two cases: (1) any of w_0, w_2, w_3 is sublinear in N (i.e., any of a, b, c is < 1); (2) all of them are linear in N (i.e., $a = b = c = 1$). We summarize the differences from [AR09] below before presenting the proofs:

– There are multiple sub-cases for Case (1) and the proofs of some sub-cases were omitted in [AR09]. We provide full proofs for all sub-cases.
– We observe that our fine-grained analysis is crucial to obtain good distance due to some sub-cases. In particular, we are able to show that the failure probability is negligible for any $\delta \le \frac{1}{2}$ when the ranges of i, j are $[0, w_0]$ in Equation (15). In [AR09], i, j can be as large as $\lfloor \frac{w_1}{2} \rfloor$ and $\lfloor \frac{w_2}{2} \rfloor$, respectively, and the provable bound is smaller than 1 only for small δ (e.g., $\delta < 0.2$).
– The same summand in Equation (15) was used for all messages with nonzero weights to show that the probability converges to 0 when $N \to \infty$ in [AR09]. However, this does not yield good concrete numbers as the probability could be inverse polynomial in N with poor parameters. We show that by restricting $w_0 > \frac{n}{c' \log^2 n}$ for high-weight messages, the failure probability is negligible.

4.1 Sublinear Case

For clarity, define the following

$$S(i, j, w_0, w_2, w_3) := \binom{n}{w_0} \frac{\binom{N-w_2}{i}\binom{w_2-1}{i-1}\binom{w_2-i}{w_1-2i}}{\binom{N}{w_1}} \frac{\binom{N-w_3}{j}\binom{w_3-1}{j-1}\binom{w_3-j}{w_2-2j}}{\binom{N}{w_2}}$$

Lemma 6. *Assume at least one of w_0, w_2, w_3 is sublinear in N, then the summand term $S(i, j, w_0, w_2, w_3) = \exp(-O(N/\log N))$.*

Proof. As previously stated, we can write

$$w_0 = \alpha N^a \qquad w_2 = \beta N^b \qquad w_3 = \rho N^c \qquad i = \gamma_1 N^{e_1} \qquad j = \gamma_2 N^{e_2}$$

where $1 - \frac{2 \log \log N}{\log N} \le a \le b \le c \le 1$ and $e_1, e_2 \le a$. When $c = 12$, $\rho \le \delta \le \frac{1}{2}$.

Using the bound from [KZKC08, AR09], we can bound each binomial coefficient in the summand as:

$$\left(\frac{N}{w}\right)^w \frac{1}{\varphi_N(w)} \le \binom{N}{w} \le \left(\frac{N}{w}\right)^w \varphi_w(w),$$

where $\varphi_N(w) = \exp\left(\frac{w(w-1)}{2N}\right)$. Following the proof in [AR09], we will consider the summand as the product of two terms, corresponding to each accumulation. The first accumulation, which we will denote as A_1, can be bounded as follows:

$$A_1 := \binom{n}{w_0} \frac{\binom{N-w_2}{i}\binom{w_2-1}{i-1}\binom{w_2-i}{w_1-2i}}{\binom{N}{w_1}} \leq$$

$$\left(\frac{N}{r\alpha N^a}\right)^{\alpha N^a} \left(\frac{N-\beta N^b}{\gamma_1 N^{e_1}}\right)^{\gamma_1 N^{e_1}} \left[\frac{\gamma_1 N^{e_1}}{\beta N^b}\left(\frac{\beta N^b}{\gamma_1 N^{e_1}}\right)^{\gamma_1 N^{e_1}}\right]$$

$$\left(\frac{\beta N^b - \gamma_1 N^{e_1}}{r\alpha N^a - 2\gamma_1 N^{e_1}}\right)^{r\alpha N^a - 2\gamma_1 N^{e_1}} \left(\frac{N}{r\alpha N^a}\right)^{-r\alpha N^a} \varphi_{\alpha N^a}(\alpha N^a)$$

$$(\varphi_{\gamma_1 N^{e_1}}(\gamma_1 N^{e_1}))^2 \varphi_{r\alpha N^a - 2\gamma_1 N^{e_1}}(r\alpha N^a - 2\gamma_1 N^{e_1}) \frac{1}{\varphi_N(r\alpha N^a)}$$

$$= \left(\frac{N}{r\alpha N^a}\right)^{(\alpha - r\alpha)N^a} \left(\frac{N-\beta N^b}{\gamma_1 N^{e_1}}\right)^{\gamma_1 N^{e_1}} \left(\frac{\beta N^b}{\gamma_1 N^{e_1}}\right)^{\gamma_1 N^{e_1}}$$

$$\left(\frac{\beta N^b - \gamma_1 N^{e_1}}{r\alpha N^a - 2\gamma_1 N^{e_1}}\right)^{r\alpha N^a - 2\gamma_1 N^{e_1}} \phi_{A_1}(N)$$

where $\phi_{A_1}(N)$ is the product of all the φ terms, and the last equality follows from $a \geq e_1$. Note that we can trivially write $\phi_{A_1}(N) = \exp(o(N^a \ln N))$, which is a very loose upper bound that will be used in some cases of our analysis. The second accumulation, which we will denote as A_2, can be bounded similarly:

$$A_2 := \frac{\binom{N-w_3}{j}\binom{w_3-1}{j-1}\binom{w_3-j}{w_2-2j}}{\binom{N}{w_2}} \leq$$

$$\left(\frac{N-\rho N^c}{\gamma_2 N^{e_2}}\right)^{\gamma_2 N^{e_2}} \left[\frac{\gamma_2 N^{e_2}}{\rho N^c}\left(\frac{\rho N^c}{\gamma_2 N^{e_2}}\right)^{\gamma_2 N^{e_2}}\right] \left(\frac{N}{\beta N^b}\right)^{-\beta N^b}$$

$$\left(\frac{\rho N^c - \gamma_2 N^{e_2}}{\beta N^b - 2\gamma_2 N^{e_2}}\right)^{\beta N^b - 2\gamma_2 N^{e_2}} (\varphi_{\gamma_2 N^{e_2}}(\gamma_2 N^{e_2}))^2$$

$$\varphi_{\beta N^b - 2\gamma_2 N^{e_2}}(\beta N^b - 2\gamma_2 N^{e_2}) \frac{1}{\varphi_N(\beta N^b)}$$

$$= \left(\frac{N-\rho N^c}{\gamma_2 N^{e_2}}\right)^{\gamma_2 N^{e_2}} \left(\frac{\rho N^c}{\gamma_2 N^{e_2}}\right)^{\gamma_2 N^{e_2}} \left(\frac{N}{\beta N^b}\right)^{-\beta N^b}$$

$$\left(\frac{\rho N^c - \gamma_2 N^{e_2}}{\beta N^b - 2\gamma_2 N^{e_2}}\right)^{\beta N^b - 2\gamma_2 N^{e_2}} \phi_{A_2}(N)$$

Now, we must consider 5 cases corresponding to which variables are $o(N)$:

- **Case 1:** $1 - \frac{2\log\log N}{\log N} \leq a = b \leq c < 1$
- **Case 2:** $1 - \frac{2\log\log N}{\log N} \leq a = b < c \leq 1$
- **Case 3:** $1 - \frac{2\log\log N}{\log N} \leq a < b \leq c < 1$
- **Case 4:** $1 - \frac{2\log\log N}{\log N} \leq a < b < c = 1$

– **Case 5:** $1 - \frac{2 \log\log N}{\log N} \le a < b = c = 1$

We show that in all cases, $A_1 \cdot A_2 = \exp(-O(N/\log N))$. See Appendix B.1 for the full proof.

4.2 Linear Case

We are left with the case where $a = b = c$, and we have the following lemma from [AR09]:

Lemma 7 ([AR09]). *When w_0, w_2, w_3 are all linear in N, there exists a constant $0 < \delta < 1$ such that for all $w_3 \le \delta$, $S(i, j, w_0, w_2, w_3) = \exp(-O(N))$.*

We rely on another variant of Stirling's inequality shown in [BCF⁺25, Lemma 9.4]:

$$0.61664 \cdot 2^{nH(\frac{k}{n})} \le \binom{n}{k} \le 0.67352 \cdot \frac{2^{n \cdot H(\frac{k}{n})}}{\sqrt{n}},$$

where $H(x) = -x \log x - (1-x)\log(1-x)$ is the binary entropy function. The goal is to write $\binom{n}{k}$ as $\mathrm{poly}(n) \cdot 2^{n \cdot H(\frac{k}{n})}$ and the constants are not used explicitly in the derivation. By applying the bound above to $S(i, j, w_0, w_2, w_3)$, we have

$$S(i, j, w_0, w_2, w_3) \le \mathrm{poly}(N) \cdot 2^{f(\alpha,\beta,\rho,\gamma_1,\gamma_2)N},$$

where

$$\begin{aligned} &f(\alpha,\beta,\rho,\gamma_1,\gamma_2) = \\ &H(\alpha)(\frac{1}{r} - 1) - H(\beta) + (1-\beta)H(\frac{\gamma_1}{1-\beta}) + \beta H(\frac{\gamma_1}{\beta}) + (\beta - \gamma_1)H(\frac{\alpha - 2\gamma_1}{\beta - \gamma_1}) \\ &\qquad + (1-\rho)H(\frac{\gamma_2}{1-\rho}) + \rho H(\frac{\gamma_2}{\rho}) + (\rho - \gamma_2)H(\frac{\beta - 2\gamma_2}{\rho - \gamma_2}), \end{aligned} \tag{16}$$

with the boundary conditions that $r\alpha/2 \le \beta \le 2\delta$ and $\rho \le \delta$. As shown in [AR09], the maximum value of $f(\alpha,\beta,\rho,\gamma_1,\gamma_2)$ is negative over the region and the boundaries. Moreover, for possible maximum values (i.e., when the partial derivatives of f are 0), $e_1 = e_2 = 1$ and γ_1, γ_2 are constants. Therefore, $f(\alpha,\beta,\rho,\gamma_1,\gamma_2)$ is upperbounded by a negative constant in this case, and thus $S(i, j, w_0, w_2, w_3) \le \mathrm{poly}(N) \cdot 2^{-O(N)} = \exp(-O(N))$. We refer the readers to [AR09, Section VI.B] for the full proof. Moreover, as shown numerically in [RA11, Table I], for $r = 4$ and $q >> n$, δ can be close to or larger than $\frac{1}{2}$.

Combining the two cases, $S(i, j, w_0, w_2, w_3) \le \exp(-O(N/\log N))$ for high-weight messages. For the entire probability in Equation (15), the ranges of w_0, w_2, w_3, i, j in the summation are all polynomial in N, thus the total probability remains $\exp(-O(N/\log N))$, which completes the proof of Theorem 3.

Message length n	Rate $1/r$	Scheme	Distance δ	Failure probability
2^{10}	1/4	Blaze	0.2	2^{-3}
		Ours	0.5	2^{-8}
	1/6	Blaze	0.2	2^{-8}
		Ours	0.5	2^{-21}
	1/8	Blaze	0.2	2^{-13}
		Ours	0.5	2^{-34}
2^{15}	1/4	Blaze	0.2	2^{-7}
		Ours	0.5	2^{-16}
	1/6	Blaze	0.2	2^{-17}
		Ours	0.5	2^{-36}
	1/8	Blaze	0.2	2^{-26}
		Ours	0.5	2^{-59}
2^{20}	1/4	Blaze	0.2	2^{-10}
		Ours	0.5	2^{-23}
	1/6	Blaze	0.2	2^{-25}
		Ours	0.5	2^{-53}
	1/8	Blaze	0.2	2^{-39}
		Ours	0.5	2^{-80}

Table 3: Distance and failure probability of our analysis and Blaze [BCF+25]. Blaze is over $\mathbb{F}_2$ and ours is over the BabyBear field with the 31-bit prime $q = 15 \cdot 2^{27} + 1$.

5 Evaluations

5.1 Distance of RAA Codes

Settings. In this section, we present the empirical evaluations of the relative distance and the failure probability of RAA codes. We implemented Sage scripts to calculate our upperbounds with different parameters, as well as the upperbounds in [BCF+25] for the binary case. The scripts were developed based on the implementations of [BFK+24] and [BCF+25].

For our bounds, we focus on the low-weight case in Section 3, as the failure probability is negligible for high-weight messages. The scripts implement the accurate calculation of the failure probability Equation (6)+Equation (9), including the $O(\frac{1}{q-1})$ terms. For example, the full formula of Equation (6) is:

$$\sum_{w_0=1}^{n/c' \log^2 n} \sum_{w_2=cw_0 \log N}^{2\delta N} \sum_{w_3=\lceil w_2/2 \rceil}^{\delta N} \binom{n}{w_0} \Bigg(\sum_{i=w_2-2w_0-1}^{w_2-1} \frac{\binom{w_3-1}{\lceil \frac{w_2-i}{2} \rceil -1}\binom{N-w_3}{\lfloor \frac{w_2-i}{2} \rfloor}\binom{w_3-\lceil \frac{w_2-i}{2} \rceil}{i}}{\binom{N}{w_2}}$$

$$+ \sum_{i=0}^{w_2-2w_0-2} \frac{\binom{w_3-1}{\lceil \frac{w_2-i}{2} \rceil -1}\binom{N-w_3}{\lfloor \frac{w_2-i}{2} \rfloor}\binom{w_3-\lceil \frac{w_2-i}{2} \rceil}{i}}{\binom{N}{w_2}} \frac{(q-1)^{\lceil \frac{w_2-i}{2} \rceil}(q-2)^i}{(q-1)^{w_2}} \Bigg)$$

For Blaze, we drop the negligible term as well (Stage 2 in their case) and implement [BCF+25, Equation (8)]:

$$\sum_{w_0=1}^{2N^\gamma/r} \binom{n}{w_0} \sum_{w_2=\lceil rw_0/2 \rceil}^{\lfloor N^\gamma \rfloor} \sum_{w_3=\lceil w_2/2 \rceil}^{\delta N} \frac{\binom{rw_0}{rw_0/2}\binom{N-rw_0}{w_2-rw_0/2}}{\binom{N}{w_2}} \frac{\binom{w_2}{\lceil \frac{w_2}{2} \rceil}\binom{N-w_2}{w_3-\lceil \frac{w_2}{2} \rceil}}{\binom{N}{w_3}} \frac{rw_0/2 \cdot \lceil w_2/2 \rceil}{w_2 w_3},$$

with $\gamma = 0.36$ as in their implementation.

Unfortunately, for large n, due to the nested summations, the scripts run out of memory after long time of executions. Therefore, for $n = 2^{15}$ and 2^{20} in the tables, we use the upperbound Equation (8)+Equation (11) and drop the $O(\frac{1}{q-1})$ terms for our case. For Blaze, we implement [BCF+25, Equation (9)] as an upperbound.

Comparison with Blaze. The empirical results are shown in Table 3 for different values of n and r. For the smallest instance of $n = 2^{10}$ and $r = 4$, the distance of RAA codes over the binary field is only 0.2, with a failure probability of 2^{-3}. With our analysis, over a 31-bit prime field, the distance increases to 0.5 with a failure probability of 2^{-8}. The small instance of $n = 2^{10}$ already appears in some applications. For example, for polynomial commitments based on error-correcting codes [GLS+23], the message length is only square-root in the size of the polynomial, see Section 5.2. When $r = 6$, our failure probability drops significantly to 2^{-21}, which is already considered acceptable in the literature, as this failure probability is statistical in the sampling phase of the RAA code and is not based on any cryptographic assumptions. By contrast, the failure probability of Blaze is 2^{-8}. Moreover, the failure probabilities decay dramatically when n becomes larger, which is expected as they are inverse polynomial in n. We note that the failure probability of Blaze in the table is without the additional testing on all messages of weight 1 or 2, thus the probability is higher than those reported in [BCF+25]. The additional testing can be applied to reduce the failure probability of both cases[2].

In Table 3, we use the BabyBear field with $q = 15 \cdot 2^{27+1}$ as the finite field for our case, which is the smallest prime number used as the base field in zkSNARKs [RIS23] (other than the binary extension field) to the best of our knowledge. We increased q to a 63-bit prime in our experiments, but the

[2] For RAA codes over $\mathbb{F}_q$, one does not need to try all messages with different values. It suffices to test the positions of nonzeros and see if the weight remains high after accumulation, using the IOWE of the accumulator matrix.

Message length n	Rate $1/r$	Failure probability		
		$\delta = 0.3$	$\delta = 0.4$	$\delta = 0.5$
2^{10}	1/4	2^{-10}	2^{-9}	2^{-8}
	1/6	2^{-24}	2^{-22}	2^{-21}
2^{15}	1/4	2^{-19}	2^{-17}	2^{-16}
	1/6	2^{-42}	2^{-39}	2^{-36}
2^{20}	1/4	2^{-25}	2^{-24}	2^{-23}
	1/6	2^{-57}	2^{-55}	2^{-53}

Table 4: Distance and failure probability of our analysis with different parameters. Numbers are obtained over the 31-bit prime field with $q = 15 \cdot 2^{27} + 1$.

Scheme	Distance δ	Proof Size	
		$m = 2^{20}$	$m = 2^{30}$
RAA over $\mathbb{F}_{2^\ell}$ [BCF+25]	0.2	1.26 MB	40 MB
RAA over $\mathbb{F}_q$ [AHKM25]	0.05	5.16 MB	165 MB
EA over $\mathbb{F}_q$ [BFK+24]	0.1	2.56 MB	82 MB
Ours	0.5	0.48 MB	15 MB

Table 5: Proof size of PCS with different codes and distances.

scripts returned exactly the same failure probabilities as in Table 3 under the supported precision. This means that the $O(\frac{1}{q-1})$ term is indeed negligible and the dominating part comes from the combinatorial terms. Therefore, the failure probabilities in Table 3 hold for all q with more than 31 bits.

Compared with other related works, the analysis in [AHKM25] only achieves the relative distance $\delta = 0.05$ with the failure probability of 2^{-21} for $n = 2^{15}$ and $r = 4$ over a large finite field. The analysis in [BFK+24] on the EA code achieves the relative distance of $\delta = 0.1$ with the failure probability of 2^{-100} for $n = 2^{10}$ and rate $\frac{1}{2}$. However, this distance is based on a conjecture [BFK+24, Conjecture 4.6], and the encoding time of EA code is $O(n \log n)$.

Overall, our experiments demonstrate that our provable distance is significantly better than those in prior works. In particular, we show that the distance of RAA codes over a large finite field is better than that of the binary field, with a much smaller failure probability.

Additional results. In Table 4, we provide additional results for the distance and failure probability of our analysis. This is achieved by tweaking the constant c in Equation (6) and Equation (9). Recall that the failure probability of Stage 2 in Equation (6) increases as c decreases, while the probability of Stage 1 in Equation (9) decreases. For a target provable distance δ, we adjust c to balance the failure probabilities of the two stages. As shown in the table, for the same parameters of n and r, we can achieve smaller provable distances with smaller failure probabilties.

5.2 Proof size of Polynomial Commitment based on Error-Correcting Codes

To further demonstrate the advantage of our RAA code analysis, we evaluate the performance of a polynomial commitment scheme using the RAA code. The construction of the PCS in [GLS+23] is presented in Protocol 1 of Appendix C. The RAA code over a large finite field can be used as a drop-in replacement in existing implementations such as that of [BFK+24] based on the EA code. The superior encoding time of the RAA code, and thus the good prover time of the PCS, has already been demonstrated in [BCF+25]. Therefore, we focus on the proof size of the PCS. We rely on the following theorem to calculate the proof size:

Theorem 4 ([GLS+23]). *For a polynomial of size m, Protocol 1 is a polynomial commitment scheme with soundness error $\frac{m}{q} + (1 - \frac{\delta}{3})^t + (1 - \frac{2\delta}{3})^t$, where δ is the relative distance of the code, and t is the number of opened columns in Protocol 1.*

With this theorem, to achieve the soundness error of $2^{-\lambda}$, it suffices to set $t = \lambda \cdot \frac{1}{-\log_2(1-\delta/3)}$ (assuming that q is large enough). Therefore, larger δ leads to smaller t, and thus smaller proof size of the PCS. In our evaluation, we set $\lambda = 128$, and set the dimensions of the matrix C representing the coefficients of the polynomial as $\sqrt{m} \times \sqrt{m}$. Therefore, the message length is $n = \sqrt{m}$. The proof consists of t columns and 2 messages computed as linear combinations of the rows, i.e., $(t+2)\sqrt{m}$ in total. We note that it is possible to further optimize the proof size by changing the dimensions of the coefficient matrix [GLS+23], but to clearly show the relationship between the distance and the proof size, we do not apply this optimization.

Table 5 shows the proof size of the PCS using different codes and distances. "Scheme" refers to the distance achieved by the analysis in the corresponding paper. As the analysis of [BCF+25] is for the binary field, the PCS has to be constructed over the binary extension field $\mathbb{F}_{2^\ell}$, over which the RAA code obviously inherits the distance of the base field. Our analysis applies to both $\mathbb{F}_q$ and $\mathbb{F}_{2^\ell}$ as we do not utilize any other property of the field than its size. As shown in the table, with our better distance, the proof size is improved by 2.6× compared to the PCS based on the provable distance of RAA codes in [BCF+25]. It is 5.4× smaller than the proof size of the PCS in [BFK+24] using the EA code, and 11× smaller than that using the distance achieved in [AHKM25]. As PCS is a major building block of many zkSNARK schemes, our improvement directly leads to smaller proof size of zkSNARKs using RAA codes.

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A RAA over the Binary Field

A.1 Comparison with the Binary Field in Stage 2

Recall that for $\mathbb{F}_2$, $N \in \mathbb{N}$ and $w_2, w_3 \in [N]$, we have

$$A^{N,2}_{w_2,w_3} = \binom{w_3 - 1}{\lceil \frac{w_2}{2} \rceil - 1}\binom{N - w_3}{\lfloor \frac{w_2}{2} \rfloor}.$$

Similar to Lemma 4, for all messages $\mathbf{x}$ with weight $0 < w_0 \leq \frac{n}{c' \log^2 n}$, suppose the weight $\mathsf{wt}(\mathbf{xM}) = w_2 > cw_0 \log N$, then by the union bound,

$$\begin{aligned}
&\Pr_{\mathbf{P}_2}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{xMP}_2\mathbf{A}) \leq \delta N] \\
&\leq \sum_{w_0=1}^{n/c' \log^2 n} \sum_{w_2=cw_0 \log N+1}^{N} \sum_{w_3=1}^{\delta N} \Pr_{\mathbf{P}_2}[\exists \mathbf{x} \text{ s.t. } \mathsf{wt}(\mathbf{xMP}_2\mathbf{A}) = w_3 | \mathsf{wt}(\mathbf{x}) = w_0, \mathsf{wt}(\mathbf{xM}) = w_2] \\
&= \sum_{w_0=1}^{n/c' \log^2 n} \binom{n}{w_0} \sum_{w_2=cw_0 \log N+1}^{N} \sum_{w_3=1}^{\delta N} \frac{A^{N,2}_{w_2,w_3}}{\binom{N}{w_2}},
\end{aligned}$$

Where $\mathbf{P}_2$ is just a permutation matrix in the binary case. As the ranges of w_0, w_2 are the same, we drop the summations and $\binom{n}{w_0}$ and focus on comparing

$$\sum_{w_3=1}^{\delta N} \sum_{i=w_2-2w_0-1}^{w_2-1} \frac{\binom{w_3-1}{\lceil \frac{w_2-i}{2} \rceil - 1}\binom{N-w_3}{\lfloor \frac{w_2-i}{2} \rfloor}\binom{w_3 - \lceil \frac{w_2-i}{2} \rceil}{i}}{\binom{N}{w_2}} \tag{17}$$

for the finite field case derived from the proof of Lemma 4 with

$$\sum_{w_3=1}^{\delta N} \frac{\binom{w_3-1}{\lceil \frac{w_2}{2} \rceil - 1}\binom{N-w_3}{\lfloor \frac{w_2}{2} \rfloor}}{\binom{N}{w_2}} \tag{18}$$

for the binary case.

As shown in the proof of Lemma 4, Equation (17) is

$$\begin{aligned}
&\leq (2w_0 + 1)(w_2 - 2w_0)\left(\frac{e\delta N}{w_0}\right)^{w_0}\left(\frac{e(1-\delta)N}{w_0}\right)^{w_0}\delta^{w_2} \\
&\leq \frac{ce^{2w_0}(2 + 1/w_0)}{w_0^{2w_0-2}}(1-\delta)^{w_0} N^{w_0(2+c\log(\delta))} \log N. \qquad (19)
\end{aligned}$$

when $cw_0 \log N < w_2 \leq \delta N/2$. The probability is negligible when $w_2 > \delta N/2$ and we omit that case for simplicity.

As shown in the proof of [BFK+24, Lemma 3.6], Equation (18) is maximized at $w_3 = \delta N$ for $\delta < 1/2$ as well, and is upper-bounded by

$$\begin{aligned}
&\leq \left(4e^2\delta(1-\delta)\right)^{\lfloor w_2/2 \rfloor} \cdot e^2 \delta N \\
&\leq e^2 \delta N^{1 + \frac{cw_0 \log(4e^2\delta(1-\delta))}{2}}. \qquad (20)
\end{aligned}$$

In order for the probability to be less than 1, $4e^2\delta(1-\delta) < 1$ and thus the provable distance δ must be significantly smaller than $1/2$. Moreover, the upper-bound in Equation (19) above is significantly smaller than the upper-bound in Equation (20). To see this, we start with $w_0 = 1$, which is the dominating term in both cases. Equation (19) is $\tilde{O}(N^{2+c\log\delta})$, while Equation (20) is $O(N^{1+\frac{c\log(4e^2\delta(1-\delta))}{2}})$. As $\delta < 1/2$, $\delta < 1-\delta$, thus $N^{1+\frac{c\log(4e^2\delta(1-\delta))}{2}} > N^{1+c(1+\log e+\log\delta)} > N^{2+c\log\delta}$ for any $c \geq 1$. In fact, the inequalities are quite loose. For example, when $\delta = 0.02$ and $c = 6$, the bound of the binary case is only $O(N^{-1.37})$, while the bound of the finite field case is $\tilde{O}(N^{-31.8})$.

For $w_0 > 1$, observe that Equation (19) decays much faster with w_0 than Equation (20). Therefore, the provable upperbound of the failure probability of Stage 2 over a large finite field is significantly smaller than the provable upperbound over the binary field.

A.2 Amplification of RA over the Binary Field

In this section, we want to prove that $w_2 \geq cw_0 \log n$ for some constant c with high probability over $\mathbb{F}_2$ for low-weight messages (i.e. when $w_0 = O(n/\log^2 n)$). In other words, we want to show the following probability is small:

$$
\begin{aligned}
&\Pr_{\mathbf{\Pi}}[\exists \mathbf{x} \neq 0 \text{ s.t. } \mathsf{wt}(\mathbf{x}\mathbf{R}\mathbf{\Pi}\mathbf{A}) \leq cw_0 \log n | \mathsf{wt}(\mathbf{x}) \leq c'n/\log^2 n] \\
&\quad = \sum_{w_0=1}^{c'n/\log^2 n} \binom{n}{w_0} \Pr_{\mathbf{\Pi}}[\mathsf{wt}(\mathbf{x}\mathbf{R}\mathbf{\Pi}\mathbf{A}) \leq cw_0 \log n | \mathsf{wt}(\mathbf{x}) = w_0] \\
&\quad \leq \sum_{w_0=1}^{c'n/\log^2 n} \binom{n}{w_0} \sum_{w_2=0}^{cw_0 \log n} \Pr_{\mathbf{\Pi}}[\mathsf{wt}(\mathbf{x}\mathbf{R}\mathbf{\Pi}\mathbf{A}) = w_2 | \mathsf{wt}(\mathbf{x}) = w_0] \\
&\quad = \sum_{w_0=1}^{c'n/\log^2 n} \binom{n}{w_0} \sum_{w_2=0}^{cw_0 \log n} \frac{\binom{w_2-1}{\lceil \frac{rw_0}{2} \rceil - 1}\binom{rn-w_2}{\lfloor \frac{rw_0}{2} \rfloor}}{\binom{rn}{rw_0}}
\end{aligned}
$$

where the third line follows from the union bound and the last line follows from uniformity of $\mathbf{\Pi}$.

For simplicity, we will assume that $r = 2k$ for some integer k. We will approximate the innermost sum using Stirling's Approximation $(\frac{n}{k})^k \leq \binom{n}{k} \leq (\frac{en}{k})^k$.

$$
\begin{aligned}
&\leq \sum_{w_2=kw_0}^{cw_0 \log n} \left(\frac{e(w_2-1)}{kw_0-1}\right)^{kw_0-1} \left(\frac{e(rn-w_2)}{kw_0}\right)^{kw_0} \left(\frac{rw_0}{rn}\right)^{rw_0} \\
&\leq \sum_{w_2=kw_0}^{cw_0 \log n} \left(\frac{e^2(w_2-1)(rn-w_2)}{kw_0(kw_0-1)}\right)^{kw_0} \left(\frac{w_0}{n}\right)^{2kw_0} \\
&= \left(\frac{e^2 w_0}{n^2 k(kw_0-1)}\right)^{kw_0} \sum_{w_2=kw_0}^{cw_0 \log n} ((rn-w_2)(w_2-1))^{kw_0}
\end{aligned}
$$

Now we will compute the maximum of this sum given that $w_0 \in (c, n/\log^2 n)$. Notice that the summand is a quadratic of the form $f(w_2) = -w_2^2 + (rn+1)w_2 - rn$, so the maximum is at $w_2 = \frac{rn+1}{2}$. But recall that $w_2 \in [kw_0, cw_0 \log n]$. If we assume that $w_0 \leq \frac{r}{2c}\frac{n}{logn}$, then we can say that the maximum is at $w_2 = cw_0 \log n$. Realistically this will hold for most values of c' since $c' = O(1)$ and we just need $c' \leq \frac{r}{2c} \log n$. So, we continue with this assumption:

$$
\begin{aligned}
&\leq \left(\frac{e^2 w_0}{n^2 k(kw_0 - 1)}\right)^{kw_0} \left((rn - cw_0 \log n)(cw_0 \log n - 1)\right)^{kw_0} \cdot \\
&\qquad (cw_0 \log n - kw_0 + 1) \\
&\leq \left(\frac{e^2 w_0}{n^2 k(kw_0 - 1)}\right)^{kw_0} (rn)^{kw_0} (cw_0 \log n)^{kw_0} (cw_0 \log n) \\
&\leq \left(\frac{2e^2 cw_0^2}{(kw_0 - 1)}\right)^{kw_0} cw_0 (\log n)^{kw_0+1} n^{-kw_0}
\end{aligned}
$$

Multiplying by the binomial term $\binom{n}{w_0}$ in the original sum, we have that the main summand is:

$$
\begin{aligned}
&\leq cw_0 \left(\frac{en}{w_0}\right)^{w_0} \left(\frac{2e^2 cw_0^2}{(kw_0 - 1)}\right)^{kw_0} (\log n)^{kw_0+1} n^{-kw_0} \\
&\leq cw_0 \left(\frac{e}{w_0}\right)^{w_0} \left(\frac{2e^2 cw_0^2}{(kw_0 - 1)}\right)^{kw_0} (\log n)^{kw_0+1} n^{w_0(1-k)}
\end{aligned}
$$

Finally, to upper bound the entire sum, we can first show that the above term is decreasing in w_0. To do this, we will just compute the ratio between consecutive terms.

$$
\begin{aligned}
\frac{T(w_0+1)}{T(w_0)} &= \frac{w_0+1}{w_0} \frac{e w_0^{w_0}}{(w_0+1)^{w_0+1}} \left(\frac{2e^2 c(w_0+1)^2}{kw_0 + k - 1}\right)^{k} \cdot \\
&\qquad \left[\left(\frac{w_0+1}{w_0}\right)^2 \frac{kw_0 - 1}{kw_0 + k - 1}\right]^{kw_0} (\log n)^k n^{1-k} \\
&\leq \frac{e w_0^{w_0}}{w_0 (w_0+1)^{w_0}} \left(\frac{2e^2 c(w_0+1)^2}{kw_0 + k - 1}\right)^{k} \left(\frac{w_0+1}{w_0}\right)^{2kw_0} (\log n)^k n^{1-k} \\
&\leq \frac{e}{w_0} e^{2k-1} \left(\frac{2e^2 c(w_0+1)^2}{kw_0}\right)^{k} (\log n)^k n^{1-k}
\end{aligned}
$$

where the last line follows from the identity $(w_0 + 1/w_0)^{w_0} < e$. We will apply one more loose bound to make the equation more legible: $w_0 + 1 \leq 2w_0$, so $(w_0+1)^2 \leq 4w_0^2$:

$$
\begin{aligned}
&\leq \frac{e}{w_0} e^{2k-1} \left(\frac{8e^2 cw_0}{k}\right)^{k} (\log n)^k n^{1-k} \\
&= e^{2k} \left(\frac{8e^2 c}{k}\right)^{k} w_0^{k-1} (\log n)^k n^{1-k}
\end{aligned}
$$

For large enough n, clearly the n^{1-k} will dominate and make this ratio < 1, so our terms are decreasing in w_0.

Going back to our main sum, this means the summand is maximized at $w_0 = 1$:

$$cw_0 \left(\frac{e}{w_0}\right)^{w_0} \left(\frac{2e^2 cw_0^2}{(kw_0 - 1)}\right)^{kw_0} (\log n)^{kw_0+1} n^{w_0(1-k)} \\ \leq ce \left(\frac{2e^2 c}{k-1}\right)^k \log^{k+1}(n) \frac{1}{n^{k-1}}$$

Since we have the factor of $n^{w_0(1-k)}$, terms from $w_0 = 2$ onward will be asymptotically dominated by the first term, so we can say that

$$\Pr_{\mathbf{\Pi}}[\exists \mathbf{x} \neq 0 \text{ s.t. } \mathsf{wt}(\mathbf{xR\Pi A}) \leq cw_0 \log n | w_0 \leq c'n/\log^2 n] \\ = O\left(\frac{\log^{k+1}(n)}{n^{k-1}}\right) = \tilde{O}(n^{1-\frac{r}{2}}).$$

B Additional Lemmas and Proofs

Lemma 8. *Let $w_1 \geq 4w_0$. Then, the summand $S(i, w_1, cw_0 \log N)$ is decreasing with i on the interval $i \in [(r-2)w_0 - 1, rw_0 - 1]$.*

Proof. We will split this into cases corresponding to the parity of $rw_0 - i$.
Case 1 (odd $rw_0 - i$). Let $rw_0 - i = 2m + 1$, where m is an integer in the interval $[0, w_0]$. We will consider the ratio between consecutive terms over i and show that it is strictly less than 1 as done in the corresponding lemma for δN.

$$\frac{S(i+1, w_1, cw_0 \log N)}{S(i, w_1, cw_0 \log N)} = \frac{\binom{cw_0 \log N - 1}{m-1}\binom{N - cw_0 \log N}{m}\binom{cw_0 \log N - m}{i+1}}{\binom{cw_0 \log N - 1}{m}\binom{N - cw_0 \log N}{m}\binom{cw_0 \log N - m - 1}{i}} \\ = \frac{m}{cw_0 \log N - m} \frac{cw_0 \log N - m}{i+1} = \frac{m}{rw_0 - 2m}$$

In order to have $\frac{m}{rw_0 - 2m} < 1$, we need $rw_0 > 3m$, or $r > 3$.

Case 2 (even $rw_0 - i$). Let $rw_0 - i = 2m$, where m is now in the interval $[1, w_0]$. Since we are considering low input weights, we can use the following bound:

$$w_0 \leq \frac{n}{c' \log^2 n} = \frac{N/r}{c' \log^2(N/r)} = \frac{N/r}{c'(\log N - \log r)^2)} \\ = \frac{N}{rc'(\log^2 N - 2\log N \log r - \log^2 r)} \leq \frac{N}{rc'(\log^2 N - 3\log N \log r)}$$

We will again compute the ratio between consecutive terms over i:

$$
\begin{aligned}
\frac{S(i+1,w_1,cw_0\log N)}{S(i,w_1,cw_0\log N)} &= \frac{\binom{cw_0\log N-1}{m-1}\binom{N-cw_0\log N}{m-1}\binom{cw_0\log N-m}{i+1}}{\binom{cw_0\log N-1}{m-1}\binom{N-cw_0\log N}{m}\binom{cw_0\log N-m}{i}} \\
&= \frac{m}{N-cw_0\log N-m+1}\frac{cw_0\log N-m-i}{i+1} \\
&\leq \frac{cw_0\log N}{N-cw_0\log N-w_0}\frac{w_0}{(r-2)w_0} \\
&\leq \frac{cw_0\log N}{N(1-\frac{c}{rc'(\log N-3\log r)}-\frac{1}{rc'(\log^2 N-3\log N\log r)})}\frac{1}{r-2} \\
&\leq \frac{cw_0\log N}{N/2}\frac{1}{r-2} \\
&\leq \frac{2c}{c'r(r-2)(\log N-3\log r)} \\
&\leq \frac{c}{c'r(\log N-3\log r)}
\end{aligned}
$$

The third to last inequality follows from c and r being constants. The last line follows from $r \geq 4$. Since c is a constant, we can conclude that the term is strictly less than 1 for large enough N. Thus $S(i,w_1,cw_0\log N)$ is decreasing with i on the interval $i \in [(r-2)w_0-1, rw_0-1]$. □

B.1 Full Proof of Lemma 6

Case 1: $1-\frac{2\log\log N}{\log N} \leq a = b \leq c < 1$

We will calculate the order of both terms using the fact that $a=b$ and $\phi_{A_1}(N) = \exp(o(N^a \ln N))$.

$$
\begin{aligned}
A_1 \leq{}& (O(1)N^{1-a})^{(\alpha-r\alpha)N^a}(O(1)N^{a-e_1}(N^{1-a}-\beta))^{\gamma_1 N^{e_1}}(O(1)N^{a-e_1})^{\gamma_1 N^{e_1}} \\
&\left(\frac{\beta N^{a-e_1}-\gamma_1}{r\alpha N^{a-e_1}-2\gamma_1}\right)^{r\alpha N^a-2\gamma_1 N^{e_1}}\exp(o(N^a\ln N)) \\
={}& \exp((1-a)(\alpha-r\alpha+\gamma_1^*)N^a\ln N+o(N^a\ln N))
\end{aligned}
$$

where $\gamma_1^*=\gamma_1$ when $a=e_1$ and 0 otherwise. This can be seen by considering the subcases $a=e_1$ and $a>e_1$ separately. Then, in particular, $\gamma_1^* \leq \gamma_1 \leq \alpha$ as we are considering $0 \leq i \leq w_0$. Note that the constants and lower order terms get absorbed by the $o(N^a \ln N)$. Finally, we can write

$$
A_1 \leq \exp(-(r-2)(1-a)\alpha N^a \ln N + o(N^a\ln N))
$$

Since we have $r \geq 4$, we have shown that A_1 will decay to 0 as $N \to \infty$.

We can bound A_2 similarly, using $\gamma_2^*=\gamma_2$ when $b=e_2$ and 0 otherwise, and the bound $\gamma_2^* \leq \alpha$. Because $a=b$, we also have the bound $\beta \geq r\alpha/2$, so $\gamma_2^* \leq 2\beta/r \leq \beta/2$.

$$
\begin{aligned}
A_2 &\le (O(1)N^{c-e_2}(N^{1-c}-\rho))^{\gamma_2 N^{e_2}}(O(1)N^{c-e_2})^{\gamma_2 N^{e_2}}(O(1)N^{1-b})^{-\beta N^b} \\
&\qquad (O(1)N^{c-b}+O(1))^{\beta N^b-2\gamma_2 N^{e_2}}\exp(o(N^b\ln N)) \\
&= \exp((1-c)(\gamma_2^*-\beta)N^b\ln N + o(N^b\ln N)) \\
&\le \exp(-(1-c)(\beta/2)N^b\ln N + o(N^b\ln N))
\end{aligned}
$$

where we use the fact that $(N^{c-b})^{\beta N^b} = (N^{b-c})^{-\beta N^b}$ to obtain the second equality. Since $c < 1$ and $a = b \ge 1 - \frac{2\log\log N}{\log N}$, it follows that $A_1 \cdot A_2 = \exp(-O(N^a \ln N)) = \exp(-O(N/\log N))$, which is negligible.

Case 2: $1 - \frac{2\log\log N}{\log N} \le a = b < c \le 1$

As in the previous case, we have that

$$A_1 = \exp(-(r-2)(1-a)\alpha N^a \ln N + o(N^a \ln N))$$

Since $c = 1$ and $a = b$, the bound on the second term becomes the following:

$$A_2 = \exp(o(N^a \ln N))$$

Then, in particular,

$$A_1 \cdot A_2 = \exp(-(r-2)(1-a)\alpha N^a \ln N + o(N^a \ln N))$$

Case 3: $1 - \frac{2\log\log N}{\log N} \le a < b \le c < 1$

We can derive a bound on A_1 without the assumption that $a = b$ in a similar way as before.

$$
\begin{aligned}
A_1 &\le (O(1)N^{1-a})^{(\alpha-r\alpha)N^a}(O(1)N^{b-e_1}(N^{1-b}-\beta))^{\gamma_1 N^{e_1}}(O(1)N^{b-e_1})^{\gamma_1 N^{e_1}} \\
&\qquad \left(\frac{\beta N^{b-e_1}-\gamma_1}{r\alpha N^{a-e_1}-2\gamma_1}\right)^{r\alpha N^a-2\gamma_1 N^{e_1}} \exp(o(N^a\ln N)) \\
&= \exp([(1-a)(\alpha-r\alpha+\gamma_1^*)+(b-a)r\alpha]N^a\ln N + o(N^a\ln N)) \\
&\le \exp(\alpha[-(1-a)(r-2)+(b-a)r]N^a\ln N + o(N^a\ln N))
\end{aligned}
$$

The derivation for A_2 remains the same as in the first case.

$$A_2 = \exp(-(1-c)(\beta/2)N^b \ln N + o(N^b \ln N))$$

Because $b > a$, we have that

$$A_1 \cdot A_2 = \exp(-(1-c)(\beta/2)N^b \ln N + o(N^b \ln N))$$

Case 4: $1 - \frac{2\log\log N}{\log N} \le a < b < c = 1$

The bound on A_1 remains the same as in the previous case. To bound A_2, we will now need to consider the coefficients.

$$A_2 \le \left(\frac{1-\rho}{\gamma_2}N^{1-e_2}\right)^{\gamma_2 N^{e_2}} \left(\frac{\rho}{\gamma_2}N^{1-e_2}\right)^{\gamma_2 N^{e_2}} \left(\frac{1}{\beta}N^{1-b}\right)^{-\beta N^b}$$
$$\left(\left(\frac{\rho-o(1)}{\beta-o(1)}\right)N^{1-b}\right)^{\beta N^b - 2\gamma_2 N^{e_2}} \cdot \phi_{A_2}(N)$$

where

$$\begin{aligned}
\phi_{A_2}(N) &= \frac{(\varphi_{\gamma_2 N^{e_2}}(\gamma_2 N^{e_2}))^2 \varphi_{\beta N^b - 2\gamma_2 N^{e_2}}(\beta N^b - 2\gamma_2 N^{e_2})}{\varphi_N(\beta N^b)} \\
&= \exp\big((\gamma_2 N^{e_2} - 1) + (\beta N^b - 2\gamma_2 N^{e_2} - 1)/2 - \beta N^{b-1}(\beta N^b - 1)/2\big) \\
&\le \exp\big((\beta N^b - \beta^2 N^{2b-1} + \beta N^{b-1})/2\big) \\
&\le \exp\big(\beta N^b/2 + o(1)\big)
\end{aligned}$$

where the last line follows from $b \le 1$.

Since $a < b$, we have $e_2 < b$. Then, the first two terms in the product are $\exp(o(N^b))$.

$$\begin{aligned}
A_2 = \exp\big(&(\beta N^b - 2\gamma_2 N^{e_2})\left[(1-b)\ln N + \ln(\rho/\beta) + o(1)\right] \\
&- \beta N^b\left[(1-b)\ln N - \ln\beta\right] + \beta N^b/2 + o(N^b)\big) \\
= \exp(&(1/2 + \ln(\rho))\beta N^b + o(N^b))
\end{aligned}$$

Note that because $\rho \le \delta \le \frac{1}{2}$, the exponent is negative, thus $A_1 \cdot A_2$ is still $\exp(-O(N/\log N))$.

Case 5: $1 - \frac{2\log\log N}{\log N} \le a < b = c = 1$

This case is the same as the previous one, except we have

$$\begin{aligned}
\phi_{A_2}(N) &= \frac{(\varphi_{\gamma_2 N^{e_2}}(\gamma_2 N^{e_2}))^2 \varphi_{\beta N - 2\gamma_2 N^{e_2}}(\beta N - 2\gamma_2 N^{e_2})}{\varphi_N(\beta N)} \\
&= \exp\big((\gamma_2 N^{e_2} - 1) + (\beta N - 2\gamma_2 N^{e_2} - 1)/2 - \beta(\beta N - 1)/2\big) \\
&\le \exp\big((1-\beta)\beta N/2\big)
\end{aligned}$$

so our bound becomes

$$A_2 = \exp(((1-\beta)/2 + \ln(\rho))\beta N + o(N))$$

□

C Polynomial Commitment based on Error-Correcting Codes

In this section, we present the construction of the polynomial commitment scheme (PCS) we used in our evaluations. The description mostly follows the prior works of [GLS+23, BFK+24] verbatim.

Protocol 1 Polynomial Commitment Scheme [GLS+23]

Public input: The evaluation point $\mathbf{x}$, parsed as a tensor product $\mathbf{x} = \mathbf{x}_0 \otimes \mathbf{x}_1$;
Private input: The polynomial ϕ, the coefficients of ϕ are denoted by c.
Let C be the $[N, n, \delta N]$-linear code, $E_C : \mathbb{F}_q^n \to \mathbb{F}_q^N$ be the encoding function, $m = n \times n$. If m is not a perfect square, we can pad it to the next perfect square.
We use a python style notation to select the i-th column of a matrix $\mathsf{mat}[:, i]$.

function PC.COMMIT(ϕ)
    Parse c as a $n \times n$ matrix C. The prover computes the encoding C_1 locally. In particular, C_1 is a $n \times N$ matrix by encoding each row of the matrix C using E_C.
    for $i \in [N]$ **do**
        Compute the Merkle tree root $\mathsf{Root}_i = \mathsf{Merkle.Commit}(\mathsf{C}_1[:, \mathsf{i}])$.
    Compute a Merkle tree root $\mathsf{Root} = \mathsf{Merkle.Commit}([\mathsf{Root}_0, ..., \mathsf{Root}_{N-1}])$ and output Root as the commitment.
function PC.OPEN($\phi, \mathbf{x}$)
    $\mathcal{P}$ receives a random vector $\mathbf{r} \in \mathbb{F}_q^n$ from $\mathcal{V}$, and computes $\mathbf{y}_r = \sum_{i=0}^{n-1} \mathbf{r}[i]\mathsf{C}[i]$, $\mathbf{y}_1 = \sum_{i=0}^{n-1} \mathbf{x}_0[i]\mathsf{C}[i]$.
    $\mathcal{P}$ sends $\mathbf{y}_r, \mathbf{y}_1$ to $\mathcal{V}$.
    $\mathcal{V}$ randomly samples $t \in [N]$ indices as an array I and sends it to $\mathcal{P}$.
    for $i \in I$ **do**
        $\mathcal{P}$ sends $\mathsf{C}_1[:, i]$ and $(\mathsf{Root}_i, \pi_i) \leftarrow \mathsf{Merkle.Open}([\mathsf{Root}_0, ..., \mathsf{Root}_{N-1}], i)$ to $\mathcal{V}$.
function PC.VERIFY($\pi_{\mathbf{x}}, \mathbf{x}, y = \phi(\mathbf{x}), \mathsf{Root}$)
    Compute $\mathbf{c}_r = E_C(\mathbf{y}_r)$ and $\mathbf{c}_1 = E_C(y_1)$.
    for $\forall i \in I$ **do**
        Check if $\mathbf{c}_r[i] == \langle \mathbf{r}, \mathsf{C}_1[:, i] \rangle$.
        Check if $\mathbf{c}_1[i] == \langle \mathbf{x}_0, \mathsf{C}_1[:, i] \rangle$.
        Compute $\mathsf{Root}_i = \mathsf{Merkle.Commit}\mathsf{C}_1[:, i]$); run $\mathsf{Merkle.Verify}(\pi_i, \mathsf{Root}_i, \mathsf{Root})$.
    Check if $y == \langle \mathbf{x}_1, \mathbf{y}_1 \rangle$.
    Output 1 if all if the above conditions hold, otherwise, output 0.

A PCS allows the prover to commit to a polynomial, and later prove to the verifier that its evaluation at a particular point is computed correctly. It consists of the following algorithms:

- $\mathsf{pp} \leftarrow \mathsf{PC.setup}(1^\lambda, \mathcal{F})$: Given the security parameter and a family of polynomials, output the public parameters pp.
- $\mathsf{com} \leftarrow \mathsf{PC.commit}(\phi, \mathsf{pp})$: On input a polynomial ϕ and the public parameter pp, output the commitment $\mathcal{C}$.
- $(y, \pi) \leftarrow \mathsf{PC.open}(\phi, \mathsf{pp}, x)$: On input the polynomial, the public parameters, and the point to be evaluated at, output $y = \phi(x)$, and the proof.
- $\{0, 1\} \leftarrow \mathsf{PC.verify}(\mathsf{pp}, \mathsf{com}, x, y, \pi)$: Verify the proof against the claim $\phi(x) = y$, output 1 if verification passes, otherwise 0.

A PCS scheme satisfies the following two properties: (1) Completeness: the verification always outputs 1 if the evaluation and the proof are computed honestly; (2) Soundness: for any probabilistic polynomial time (PPT) adversary generating an incorrect evaluation, the probability of the verification outputting 1 is negligible. See [GLS+23] for the formal definitions.

Construction. We present the construction of PCS proposed in [GLS+23] based on any error-correcting code with constant relative distance. The construction uses the Merkle tree [Mer87], a data structure that allows one to commit to a vector of messages by a single hash value h, and later open any message a_i with a proof of correctness. We denote the algorithm as:

- $h \leftarrow \mathsf{Merkle.Commit}(\mathbf{a})$.
- $(a_i, \pi_i) \leftarrow \mathsf{Merkle.Open}(\mathbf{a}, i)$.
- $\{0,1\} \leftarrow \mathsf{Merkle.Verify}(\pi_\mathsf{i}, a_i, h)$.

A polynomial evaluation can be expressed as a tensor product. Here we use a multilinear polynomial as an example, but the construction works for univariate polynomials as well. Given a multilinear polynomial ϕ, its evaluation on input vector $x_0, x_1, ..., x_{\log m-1}$ is:

$$\phi(x_0, x_1, ..., x_{\log m-1}) = \sum_{i_0=0}^{1}\sum_{i_1=0}^{1}...\sum_{i_{\log N-1}=0}^{1} c_{i_0 i_1 ... i_{\log m-1}} x_0^{i_0} x_1^{i_1} ... x_{\log m-1}^{i_{\log m-1}}.$$

The degree of each variable is either 0 or 1 by the definition of a multilinear polynomial, and thus there are m monomials and coefficients with $\log m$ variables. Let $i_0 i_1 ... i_{\log m-1}$ be the binary representation of number i, i.e., $i = \sum_{j=0}^{\log m-1} 2^j i_j$. We use c to denote the coefficients where $c[i] = c_{i_0 i_1 ... i_{\log m-1}}$. Similarly, we define $X_i = x_0^{i_0} x_1^{i_1} ... x_{\log m-1}^{i_{\log m-1}}$. For $n = \sqrt{m}$, define vectors $\mathbf{x}_0, \mathbf{x}_1$ as $\mathbf{x}_0 = \{X_0, X_1, ..., X_{n-1}\}, \mathbf{x}_1 = \{X_{0\times n}, X_{1\times n}, X_{2\times n}, ..., X_{(n-1)\times n}\}$. Then we have $X = \mathbf{x}_0 \otimes \mathbf{x}_1$. The polynomial evaluation is reduced to a tensor product $\phi(x_0, x_1, ..., x_{\log m-1}) = \langle c, \mathbf{x}_0 \otimes \mathbf{x}_1 \rangle$.

The polynomial commitment scheme is presented in Protocol 1, where the message length of the error-correcting code is set to $n = \sqrt{m}$, the squre-root of the size of the polynomial m.

Finally, zkSNARKs can be constructed by combining the polynomial commitment scheme various polynomial interactive oracle proof (PIOP) protocols (e.g., [GWC19, XZZ+19, Set20, GLS+23, CBBZ22]). See [Tha22] for a survey. We omit the formal descriptions in this paper.